



RESEARCH ARTICLE

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Spartina alterniflora in coastal wetlands: ecological impacts, management trade-offs, and future pathways

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Abstract

Spartina alterniflora, one of the most widespread invasive salt-marsh plants globally, has profoundly altered the structure and functioning of coastal wetlands through its influences on sediment dynamics, biodiversity, biogeochemical cycling, and ecosystem resilience. However, the ecological consequences and management implications of its invasion remain highly controversial because its impacts vary substantially across wetland types, climatic regions, and invasion stages. To clarify the development and emerging directions of *S. alterniflora* research, this study provides a systematic review of *S. alterniflora* research, highlighting advances, challenges, and prospects in this field. Bibliometric and quantitative methods assessed research trends, while synthesized ecological findings highlight *S. alterniflora*'s impacts, such as increased soil carbon sequestration and biodiversity shifts. Literature spanning 1929–2026 from the Web of Science database and Chinese-funded projects recorded in the Fun Research database (1987–2026) formed the basis of the analysis. Bibliometric analysis, keyword co-occurrence clustering, and multiple correspondence analysis were applied to examine knowledge structure, thematic evolution, and research trends. The results demonstrate that *S. alterniflora* research has evolved from early studies focusing on plant growth, invasion ecology, and salt-marsh dynamics toward increasingly interdisciplinary investigations emphasizing blue carbon, climate change interactions, ecosystem resilience, and adaptive management. Three major thematic domains were identified: invasion ecology and biodiversity responses, biogeochemical cycling and blue carbon processes, and climate-related vulnerability and adaptation research. The ecological impacts of *S. alterniflora* were found to be strongly context-dependent across salt marshes, mangrove ecosystems, and tidal flats, generating both ecosystem benefits, such as shoreline stabilization and sediment accretion, and ecological risks including biodiversity loss, habitat homogenization, and altered trophic interactions. This review further highlights substantial uncertainties in evaluating the net climatic effects of *S. alterniflora* invasion due to spatial heterogeneity, greenhouse gas offsets, microbial feedbacks, and limitations in current blue carbon accounting frameworks. Recent advances in remote sensing, unmanned aerial vehicles, machine learning, and molecular ecology are rapidly improving the capacity to monitor invasion dynamics and assess ecosystem responses across multiple spatial and temporal scales. This study emphasizes that *S. alterniflora* should not be viewed solely as either a harmful invasive species or a beneficial ecosystem engineer, but rather as a complex ecological driver embedded within coupled coastal socio-ecological systems in the Anthropocene. Future research and management should therefore adopt integrated, context-dependent, and adaptive frameworks that simultaneously consider biodiversity conservation, climate mitigation, coastal resilience, and long-term ecosystem sustainability.

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Keywords *Spartina alterniflora*, Invasive species, Blue carbon, Salt marsh, Management

1 Introduction

Spartina alterniflora (*S. alterniflora*), commonly known as Smooth Cordgrass, is widely distributed in the intertidal areas along the Atlantic coast of North America, playing a pivotal role in coastal salt marsh ecosystems (Gao et al. 2014; Z. Wu et al. 2026a, b). It was brought to many coastal areas in Europe, South Africa, Australia, and Asia with the main goal of preventing the erosion of coastal wetland areas by decreasing the intensity of tidal waves, retaining sediments, and encouraging vertical accretion (Chung 1993; Maricle & Lee 2002). *S. alterniflora* salt marshes exert both positive ecological effects, such as pollutant breakdown, and negative impacts, like disrupting food web structure and altering biodiversity and ecosystem structure (Chung 2006; Li et al. 2005; Qin et al. 1997, 1989). According to Nellemann et al. (2009), its significant subsurface production also plays a crucial role in the carbon storage known as “blue carbon” in coastal soils. However, *S. alterniflora* may become invasive easily due to its fast and competitive growth behavior, turning mudflats and other littoral ecosystems into marshes by absorbing silt and lowering tidal energy (ISSG 2020; Partridge 1987). Shorebirds had less and worse feeding habitat as a result of the loss of these littoral ecosystems (Kerr et al. 2016); also, local biodiversity is displaced (Zhang et al. 2017). Furthermore, *S. alterniflora* can choke natural sloughs and flood control channels with its enhanced sedimentation rates (ISSG 2020). These low-lying geographic features may eventually be raised by sedimentation to the same height as the marsh plain, which would result in the loss of vital drainage and ecological structure (ISSG 2020). The International Union for Conservation of Nature (IUCN) named *S. alterniflora* one of the top 100 worst invasive alien species in the world in 2000 (Lowe et al. 2000). Since it may easily hybridize with native *Spartina* species, produce copious amounts of pollen that outcompete native stigmas, and generate more fit hybrids that readily establish and spread, controlling invasive *S. alterniflora* is difficult and exceedingly costly, with management expenses in some European countries reaching hundreds of billions of euros annually (Ayres et al. 1999; Diagne et al. 2021).

Since its introduction from the US Atlantic coast in 1979 for coastal protection (Chung 1993; X. Zhang et al. 2023a, b), *S. alterniflora* has expanded rapidly across China's coastline due to its high ecological adaptability and reproductive vigor (Lu & Zhang 2013). According to the latest national survey and systematic assessments, its distribution area reached approximately 68,000 hectares by 2022 (Jiang et al. 2024), a significant increase from earlier

estimates. This proliferation has fundamentally altered primary productivity, coastal hydrodynamics, and sedimentary processes, leading to its classification as a primary invasive species in 2003 (Qing Wang et al. 2006a, b; Q. Wang et al. 2006a, b). While *S. alterniflora* provides localized benefits such as wave attenuation, its broad-scale invasion has severely compromised mariculture and native biodiversity (Chen et al. 2009; Wan et al. 2009; L. Yang et al. 2025a, b). In December 2022, China launched a three-year national initiative, the Special Action Plan for *Spartina alterniflora* Control (2022–2025) (<https://www.gov.cn/>), with an investment exceeding 6 billion CNY, aimed at eradicating *S. alterniflora*. This ambitious plan reflects a decisive shift toward proactive management of the species in response to its significant ecological impacts.

While several studies have investigated *S. alterniflora* in different regions worldwide, there remains a lack of a comprehensive overview. Despite the considerable attention *S. alterniflora* management research has received, a systematic survey of the literature to date is still necessary to identify gaps that require focused research and to gain a better understanding and improve management effectiveness. Accordingly, we conducted a systematic review of published literature and project funding related to *S. alterniflora* in coastal ecosystems. Specifically, our objectives are: a) Analyzing the main characteristics of *S. alterniflora* research articles, especially their research areas and focused content; b) Examining the research trends and evolution of hot topics related to *S. alterniflora*; c) Analyzing the cooperation status within academia and the interaction between various research directions related to *S. alterniflora*. Rather than providing a descriptive synthesis alone, this review critically examines how a century of ecological engineering, scientific research, and management interventions have shaped both the benefits and risks associated with *S. alterniflora*. We explicitly focus on management-relevant trade-offs, unintended consequences, and decision-making contexts under ongoing environmental change, thereby aligning ecological knowledge with coastal governance needs.

2 Materials and methods

As seen in the workflow (Fig. 1), we created the review framework in two primary stages: (a) systematic literature retrieval and screening, and (b) data extraction and analytical synthesis. The workflow was designed to identify the temporal evolution, knowledge structure, and emerging

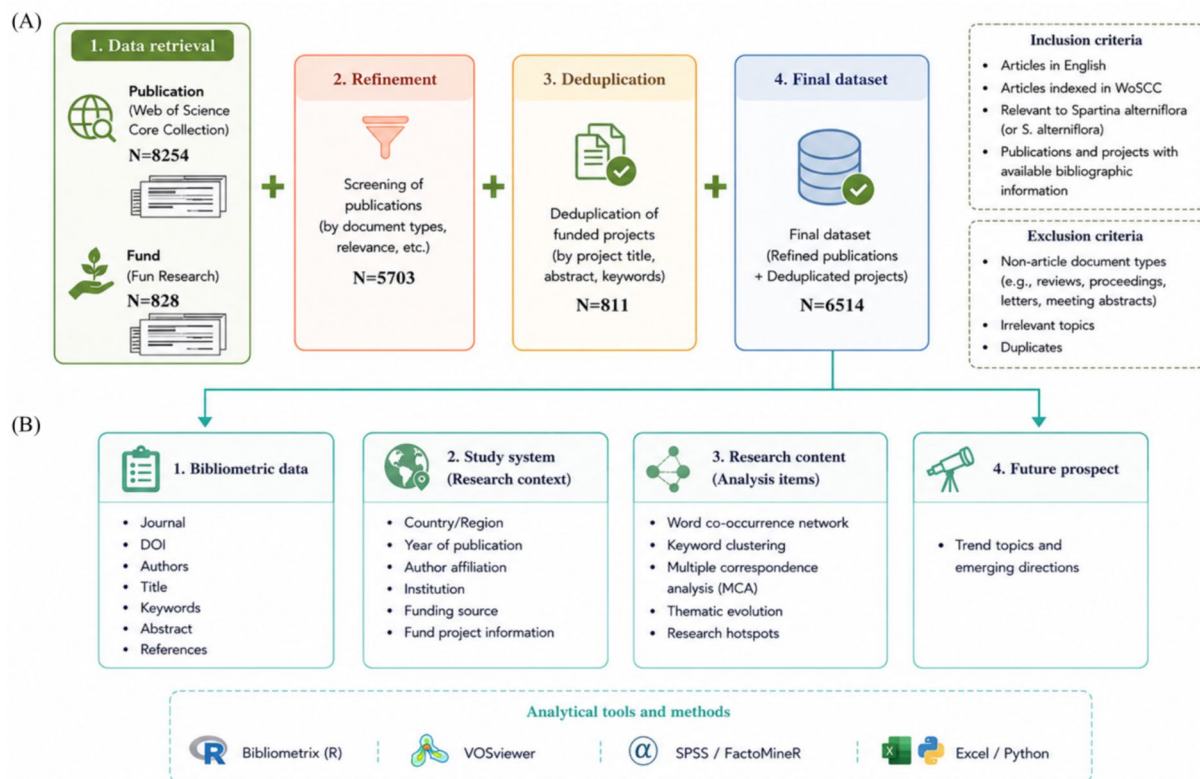


Fig. 1 **A** Diagram describing the flow of information through the different phases of the systematic literature review. **B** Summary of the data collected for the database

research directions associated with *S. alterniflora* research at both publication and funding levels.

2.1 Literature retrieval and database construction

Systematic data retrieval was conducted on April 27, 2026 using two major databases: the Web of Science (WoS) for scientific publications (<https://webofscience.clarivate.cn/>) and the Fun Research database (<http://www.funresearch.cn/>) for funded research projects in China. To ensure comprehensive coverage of the available literature, the search period spanned from 1900 to 2026. In the end, we used only the presently approved names in our analyses to construct the list of plant genera from the chosen research. For example, *Sporobolus* is the currently recognized name for *Spartina*. We did this by cross-referencing with synonyms according to <https://powo.science.kew.org/>. The search combination used was: ("*Spartina alterniflora*" OR "*Sporobolus alterniflorus*" OR "*S. alterniflora*" OR "*S. alterniflorus*"), and for the Chinese funding search: ("*Spartina alterniflora*" OR "*Sporobolus alterniflorus*" OR "*S. alterniflora*"

OR "*S. alterniflorus*") AND (NR=China). The initial search yielded 8254 publication records from WoS and 828 project records from the Fun Research database. Publication records were subsequently refined by restricting document types to research articles indexed in the Web of Science Core Collection and by excluding irrelevant records based on titles, abstracts, and keywords. This screening process resulted in 5703 eligible publication records. For the funded project database, duplicate records were identified and removed based on project titles, abstracts, and keywords, resulting in 811 unique project records. The final analytical dataset therefore consisted of 6514 records, including 5703 scientific publications and 811 funded research projects (Fig. 1A). All bibliographic records were organized and managed using Endnote 21 (Clarivate Analytics, Philadelphia, PA, USA) prior to subsequent data extraction and bibliometric analyses.

2.2 Data extraction and bibliometric analysis

Detailed bibliographic and research-related information was extracted from all records included in the final

dataset (Fig. 1B). The extracted variables included bibliometric information (e.g., journal, DOI, authors, titles, abstracts, and keywords), research system characteristics (e.g., country or region, publication year, institutional affiliation, and funding information), thematic research content associated with *S. alterniflora* studies. To characterize the knowledge structure and thematic evolution of the field, bibliometric analyses were organized into several major analytical categories, including keyword co-occurrence analysis, thematic clustering analysis, multiple correspondence analysis (MCA), and temporal trend analysis. These approaches were used to identify major research hotspots, conceptual relationships among research themes, and the long-term evolution of scientific priorities in *S. alterniflora* research. Keyword standardization was conducted prior to analysis to minimize inconsistencies caused by differences in capitalization, abbreviations, and synonymous terminology. For visualization consistency, all keywords used in the bibliometric figures were converted to lowercase. All statistical analyses and bibliometric visualizations were performed using R version 4.3.2. The primary analytical packages included: “bibliometrix” (Aria & Cuccurullo, Elsevier/2017) for bibliometric analysis and “shiny” (Chang et al. 2023) for interactive data visualization and analytical support. Additional figure editing and graphical optimization were performed using standard scientific illustration software.

3 Bibliometric patterns and evolution of research themes

3.1 Growth of research output and funding patterns

Publications on *S. alterniflora* have increased steadily since 1929, with a marked rise post-1988 (Fig. 2A). According to our literature search, the most mentioned keyword was *Spartina alterniflora*, followed by salt marsh, growth, nitrogen, vegetation, dynamics, salinity, *phragmites australis*, and carbon (Fig. 2A). China's research on *S. alterniflora* was started in 1987, and the amount of the Chinese government's investment in *S. alterniflora* research has been gradually increasing in our database (Fig. 2B). Funding rose sharply post-2015, peaking at 425.073 million CNY in 2018, driven by concerns over invasiveness and reinforced by the 2022 eradication plan, aligning research with control priorities. Comparing Fig. 2C and D, the logistic model for annual publications predicts a peak of 279 papers around 2030.2 ($R^2 = 0.899$), whereas the cumulative logistic curve gives a saturation point (K) of 12,145 papers. Together, these two projections indicate that the field of *S. alterniflora* research is entering a late growth stage, with annual output expected to plateau in the early 2030s and total publications approaching an upper limit of about 12,100.

The United States and China lead in output (Table S1), reflecting the species' native and invasive ranges, respectively. Research from the US often focuses on ecological impacts in native habitats, while Chinese studies emphasize invasion dynamics and management (e.g., Meng et al. (2020)). Journals like the Marine Ecology Progress Series and institutions such as Louisiana State University dominate (Tables S2, S3), highlighting regional research priorities. Other analyses regarding the national collaboration networks, affiliation collaborative networks, and author collaboration networks can be found in the supplementary materials (Fig. S1, S2, and S3).

3.2 Knowledge structure and thematic clustering

Although studies on *S. alterniflora* span diverse ecological and biogeochemical processes, these findings are rarely integrated into a coherent framework that informs management decisions. Since 1929, a series of milestones have been achieved in *S. alterniflora* research. Keyword co-occurrence clustering, thematic map and MCA were employed to identify the principal knowledge domains and thematic relationships within *Spartina alterniflora* research (Figs. 3, 4 and 5). The clustering analysis and thematic map group keywords based on their co-occurrence frequency within the same publications, thereby reflecting the conceptual structure and major research directions of the field. Meanwhile, the MCA ordination summarizes the relationships among high-frequency keywords in a low-dimensional space, where shorter distances between terms indicate stronger thematic associations and more frequent co-occurrence in the literature. The clustering results and thematic map (Figs. 3 and 4) reveal three major and partially interconnected research domains that collectively characterize the current knowledge structure of *S. alterniflora* studies.

Cluster 1: invasion ecology and biodiversity responses. The first cluster primarily represents invasion ecology and biodiversity responses, encompassing keywords related to invasion dynamics, vegetation change, species interactions, biodiversity, microbial communities, and ecosystem structure (Figs. 3, 4, 5 and 6A). This thematic group reflects extensive research examining how *S. alterniflora* alters the composition and functioning of coastal wetland ecosystems following invasion. Studies within this cluster have focused on competitive displacement of native vegetation, shifts in benthic and microbial communities, habitat modification, and trophic interactions across salt marshes, tidal flats, and mangrove-associated wetlands. In recent years, this research domain has expanded from conventional vegetation surveys toward multi-scale ecosystem assessments integrating microbial ecology, functional diversity, and ecosystem multifunctionality. For example, increasing attention

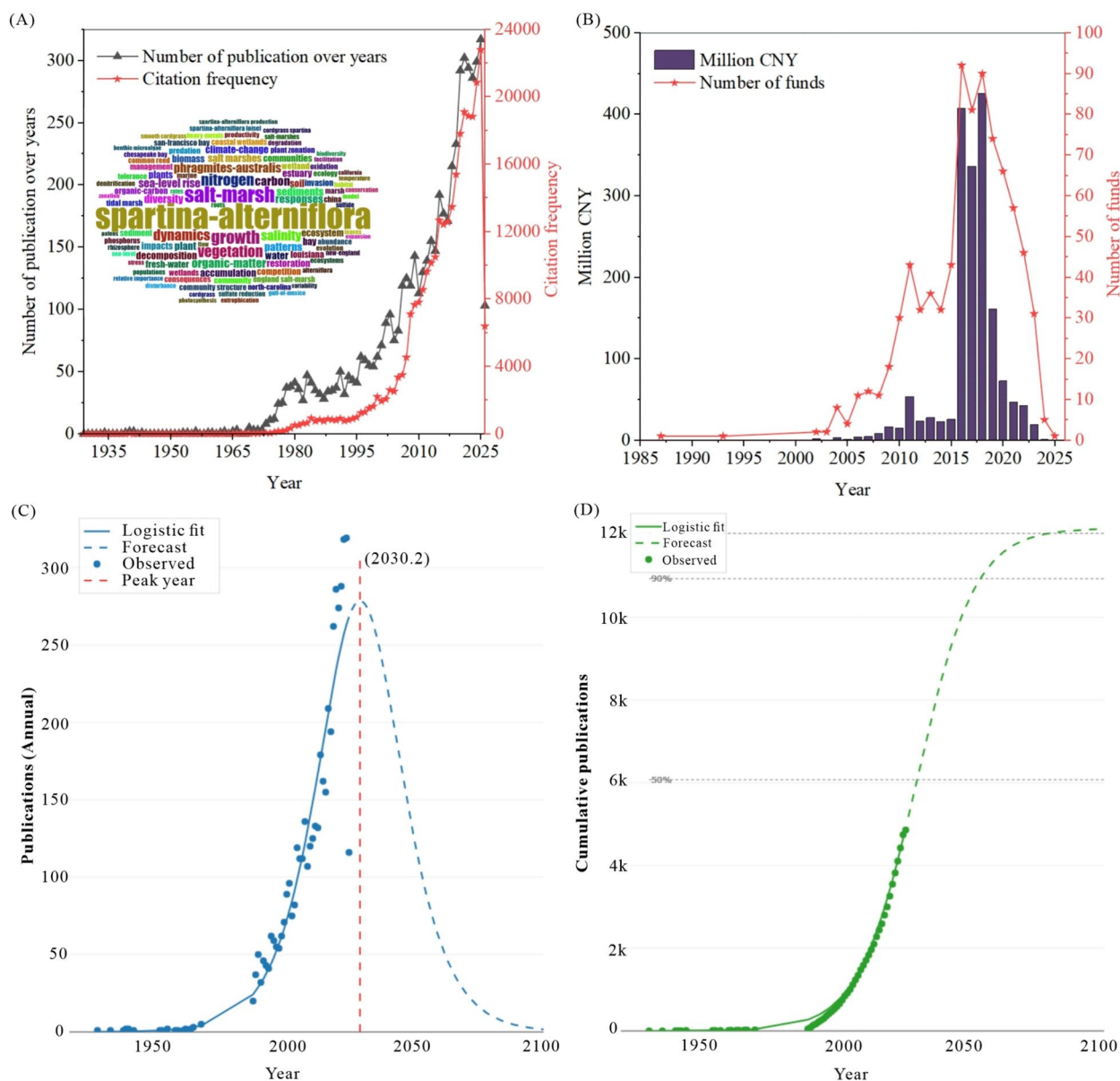


Fig. 2 **A** The number of publications and citation frequency related to *S. alterniflora* from 1929 to 2026 along with a word cloud providing the most frequently used keywords in our database. **B** Research fund of *S. alterniflora* of China from 1987 to 2026. **C** Life cycle – Annual publications, $R^2 = 0.899$, peak = 279 publications in 2030.2. **D** Cumulative growth curve, saturation (K) = 12,145 publications

has been given to changes in bacterial community composition, sulfur- and iron-cycling microorganisms, and benthic habitat suitability following *S. alterniflora* colonization (Cheng et al. 2026; Leng et al. 2025; K. Wang et al. 2004a, b, c, d, e; Y. Wang et al. 2024a, b, c, d, e). The integration of biodiversity responses with ecosystem functioning indicates a transition from species-level invasion studies toward broader socio-ecological assessments of coastal wetland resilience.

Cluster 2: biogeochemical cycling and blue carbon processes. The second cluster reflects biogeochemical cycling and blue carbon processes, including keywords such as carbon, organic matter, decomposition, accumulation, nitrogen, biomass, and salt marsh (Figs. 3, 4, 5 and 6B). This cluster highlights the central role of *S. alterniflora* in regulating coastal biogeochemical processes through its high productivity, extensive belowground biomass, and sediment-trapping capacity. Research within this domain has increasingly focused on soil organic carbon

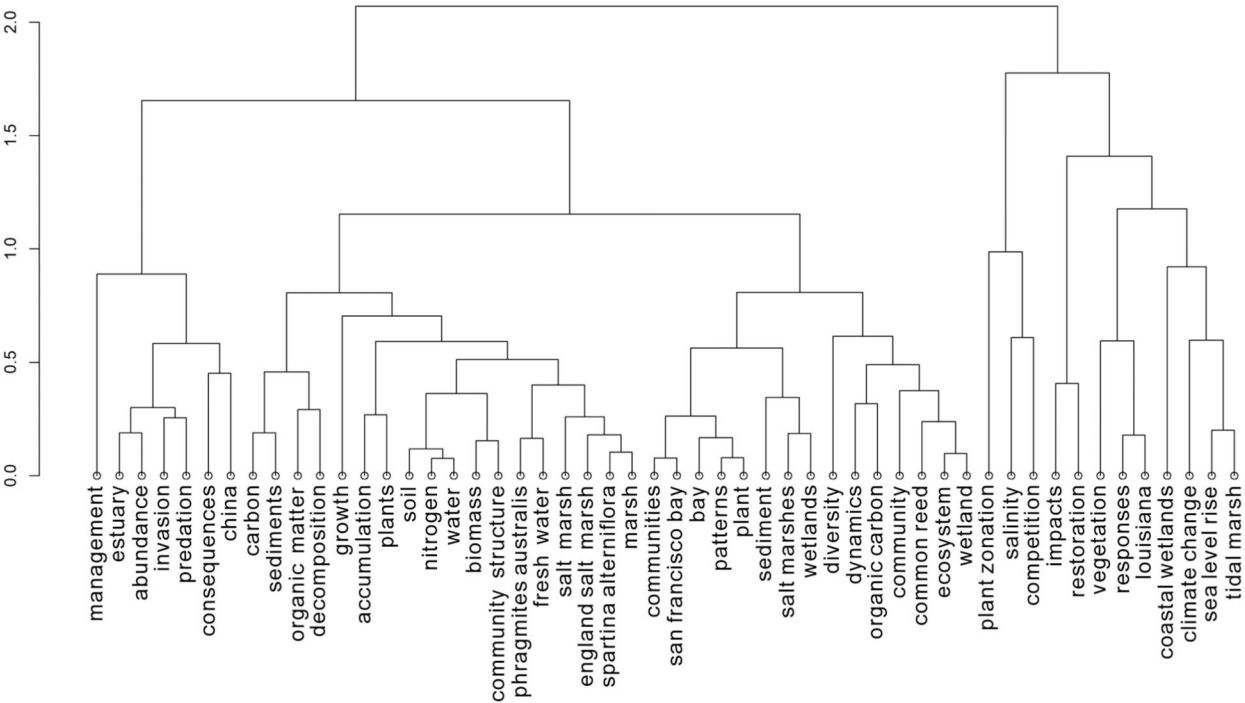


Fig. 3 Keyword cluster analysis

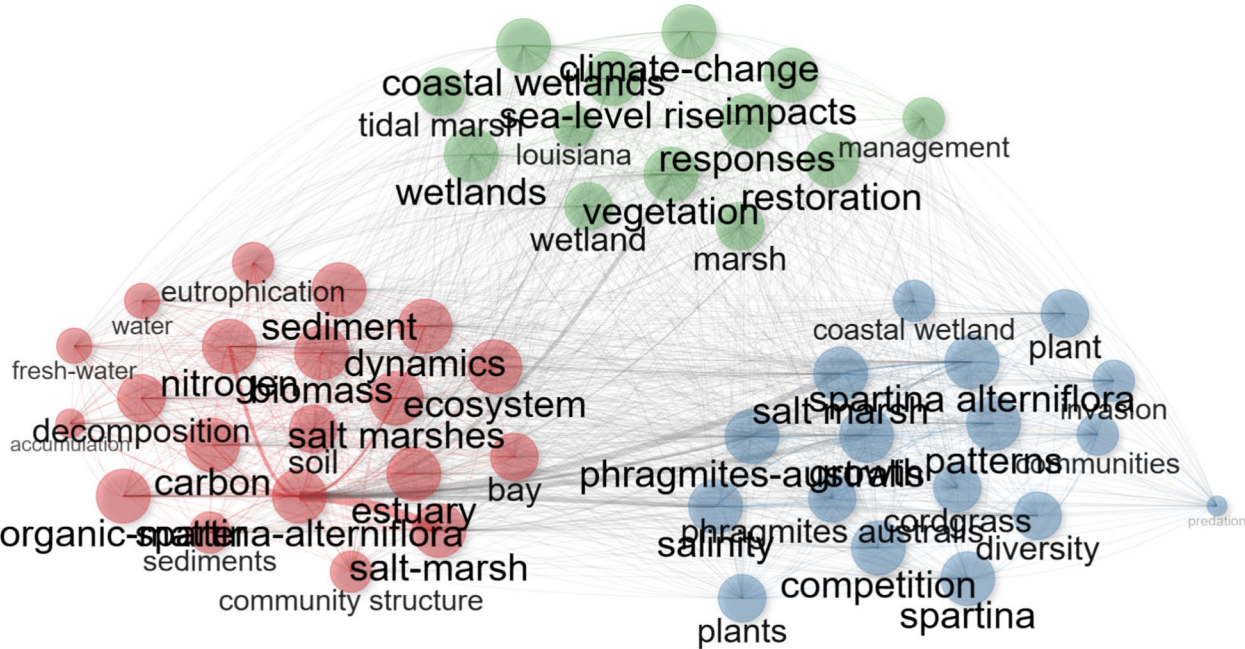
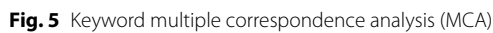


Fig. 4 Keyword thematic map

accumulation accumulation, nutrient cycling, greenhouse gas fluxes, and plant–microbe interactions. Numerous studies have demonstrated that *S. alterniflora* invasion

substantially modifies carbon and nitrogen dynamics by altering litter decomposition rates, microbial metabolism, sediment physicochemical properties, and dissolved



Cluster 3: climate-related vulnerability and adaptation research. The third cluster captures climate-related vulnerability and adaptation research, incorporating keywords such as impacts, restoration, responses, climate change, sea level rise, and habitat suitability (Figs. 3, 4, 5 and 6C). This thematic group reflects the growing recognition that *S. alterniflora* invasion is closely intertwined with global environmental change and coastal adaptation processes. Studies in this cluster have primarily investigated how climate-driven factors—including sea-level rise, warming temperatures, altered precipitation regimes, and extreme weather events—affect the distribution, expansion, and ecological consequences of *S. alterniflora*. Predictive modeling studies increasingly

To sum up, the thematic clustering analysis reveals that contemporary *S. alterniflora* research has evolved into a highly interdisciplinary field integrating invasion ecology, biogeochemistry, climate science, and coastal management. Although these thematic domains remain analytically distinguishable, substantial overlap exists among them, particularly regarding ecosystem functioning, carbon cycling, and climate adaptation. This interconnected knowledge structure underscores the necessity of evaluating *S. alterniflora* not solely as an invasive species, but as a complex ecological driver embedded within coupled coastal socio-ecological systems.

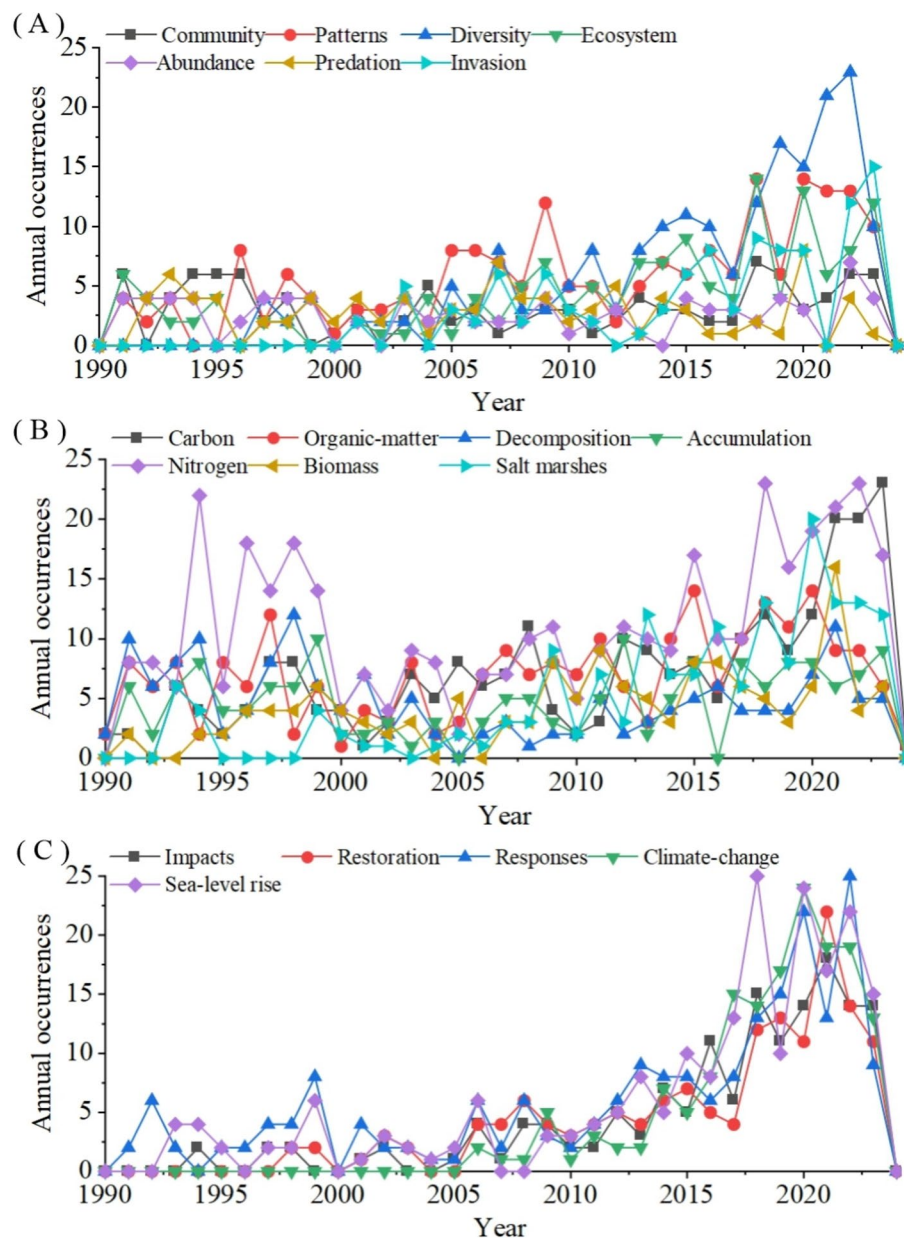


Fig. 6 Research themes of *S. alterniflora*. **A** invasion ecology and biodiversity responses, **B** biogeochemical cycling and blue carbon processes, **C** climate-related vulnerability and adaptation research

3.3 Temporal evolution of research priorities

The temporal evolution of research priorities related to *S. alterniflora* reflects not only the progression of invasion ecology and coastal wetland science, but also broader shifts in global environmental concerns, technological development, and regional management demands. High-frequency keywords identified from the bibliometric dataset ((Fig. S4, Table 1), together with emergent topic analysis (Figs. 6 and 7), reveal a clear transition in research emphasis from foundational ecological studies

toward climate-related ecosystem processes and adaptive coastal management. The high-frequency keywords that appeared in the literature included *Spartina alterniflora* (1485), salt marsh (543), growth (384), vegetation (311), nitrogen (298), and dynamics (274). The co-occurrence relationships among these keywords further demonstrate the interdisciplinary nature of *S. alterniflora* research, spanning invasion ecology, biogeochemistry, coastal geomorphology, microbial ecology, and climate adaptation studies.

Table 1 Top 15 keyword frequency statistics

Rank	Keywords	Occurrences
1	<i>Spartina alterniflora</i>	1485
2	salt marsh	543
3	growth	384
4	vegetation	311
5	nitrogen	298
6	dynamics	274
7	salinity	237
8	<i>Phragmites australis</i>	229
9	organic matter	200
10	carbon	193
11	sea level rise	173
12	diversity	172
13	patterns	169
14	responses	169
15	sediments	151

Based on the temporal distribution of trend topics (Figs. 6 and 7), the evolution of research priorities can be broadly divided into three major phases. The size of the circles associated with various keywords is proportional to their frequency in the 5703 articles, and the intricate relationships among these keywords indicate the depth and complexity of studies related to *S. alterniflora* (Fig. 7). Trend topic words are identified as hot words based on the word frequency growth rate of the titles, abstracts, and keywords.

Phase I (1991–2000): foundational ecological and physiological studies. The first phase primarily focused on the basic ecological and physiological characteristics of *S. alterniflora*, including growth dynamics, litter decomposition, salinity tolerance, and species interactions (Kneib et al. 1997). During this period, research was strongly influenced by the rapid development of invasion ecology following the conceptual frameworks proposed by Elton (1958) and subsequent ecological studies on biological invasions (Mooney & Drake 1986). High-frequency terms such as “growth,”

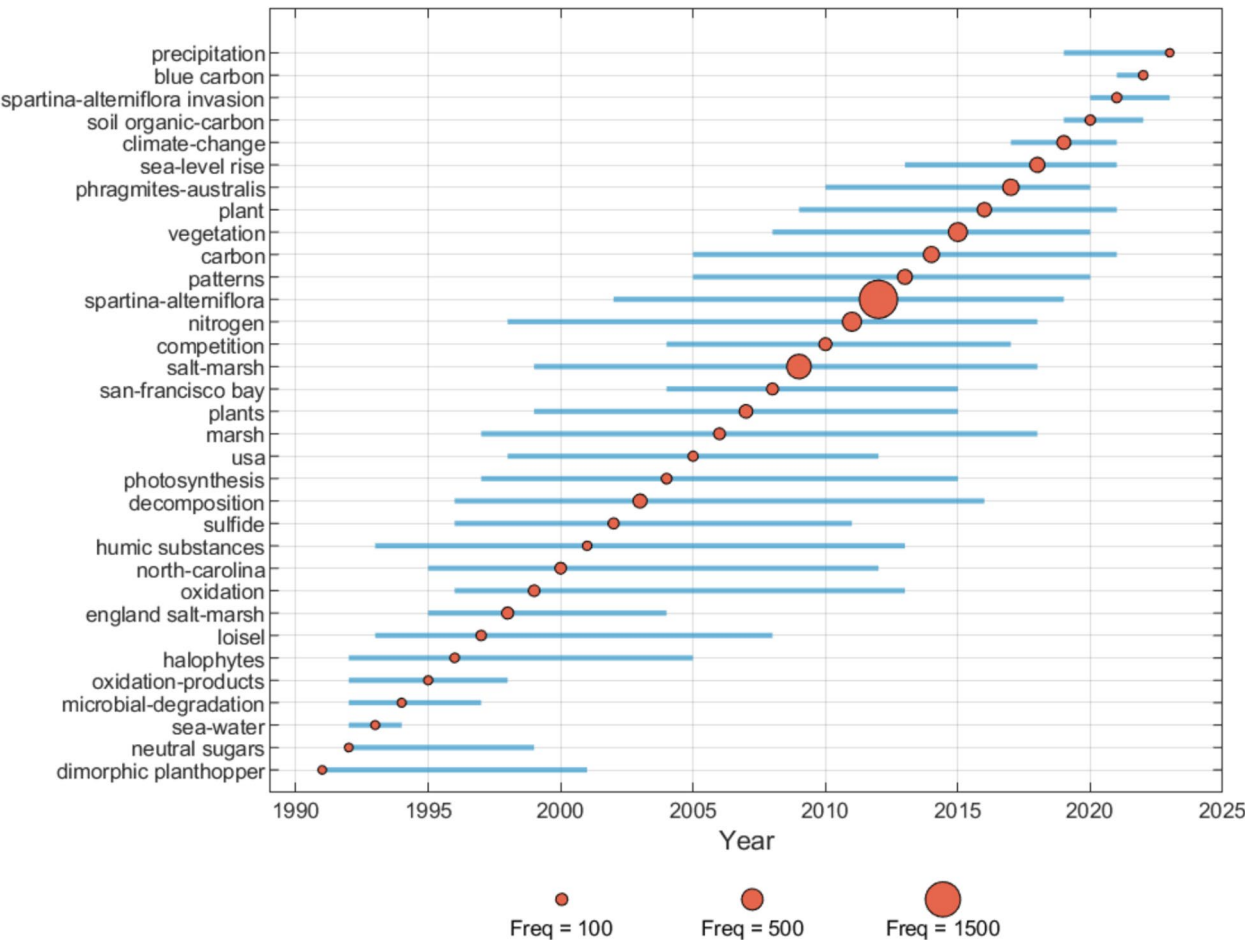


Fig. 7 Changes of the topics from 1991 to 2026

“decomposition,” “abundance,” and “salt marsh” dominated the literature, reflecting efforts to understand the mechanisms underlying the rapid expansion and competitive advantages of *S. alterniflora* in coastal wetlands (Figs. 6 and 7). This stage also coincided with the early introduction and expansion of *S. alterniflora* in East Asian coastal regions, particularly along the Chinese coast, where the species was initially valued for shoreline stabilization and sediment accretion (Qin et al. 1997). Consequently, early studies often emphasized its ecological engineering functions and its role in coastal protection, sediment trapping, and marsh formation. However, increasing observations of native vegetation displacement and habitat alteration gradually shifted attention toward its invasive characteristics and ecological risks.

Phase II (2001–2010): ecosystem processes and biogeochemical cycling. The second phase was characterized by increasing attention to ecosystem functioning and biogeochemical processes associated with *S. alterniflora* invasion. Keywords such as “nitrogen,” “organic matter,” “sediments,” “accumulation,” and “predation” became increasingly prominent during this period (Figs. 6 and 7), reflecting the growing integration of wetland biogeochemistry and ecosystem ecology into invasion research. This transition was closely associated with advances in coastal wetland science and increasing recognition of the ecological consequences of invasive species on nutrient cycling, microbial processes, and food-web dynamics. Studies increasingly examined how *S. alterniflora* altered carbon and nitrogen cycling, sediment properties, microbial community composition, and benthic ecosystem functioning (Rui-Zhao & Li-Quan 2009; Tyler et al. 2003). Simultaneously, ecological impacts on higher trophic levels, including macrobenthos and migratory birds, began to receive greater attention due to concerns over biodiversity loss and habitat degradation in estuarine wetlands (Gan et al. 2006; Zhou et al. 2009). The expansion of interdisciplinary analytical techniques during this period, including stable isotope tracing, microbial sequencing, and ecosystem-scale monitoring, further facilitated the transition from descriptive invasion studies toward process-based ecological investigations.

Phase III (2011–present): climate change, blue carbon, and adaptive management. Since approximately 2011, research priorities have increasingly shifted toward climate change interactions, blue carbon dynamics, ecosystem resilience, and management trade-offs. Emerging keywords such as “climate change,” “sea level rise,” “blue carbon,” “soil organic carbon,” “sequestration,” “habitat suitability,” and “bacterial communities” became dominant in recent years (Figs. 6 and 7), indicating a substantial broadening of research perspectives from local invasion impacts toward

global environmental change and sustainability challenges. Several drivers contributed to this transition. First, increasing international attention to climate mitigation and coastal carbon sequestration, particularly following the development of the “blue carbon” framework, stimulated extensive research on the carbon storage potential of *S. alterniflora*-dominated wetlands (J. Zhang et al. 2024a, b). Second, accelerating sea-level rise, extreme climatic events, and coastal habitat degradation heightened concerns regarding the long-term resilience of tidal marsh ecosystems under global change (J. Wang et al. 2025a, b, c, d, e). Third, large-scale ecological restoration and invasive species management programs, especially the nationwide *S. alterniflora* control initiatives implemented in China after 2022, further promoted management-oriented research on ecological trade-offs, restoration effectiveness, and adaptive governance strategies. At the same time, rapid advances in remote sensing technologies, unmanned aerial vehicles (UAVs), machine learning, and spatial modeling substantially improved the capacity to monitor invasion dynamics and evaluate ecosystem responses across broader spatial and temporal scales (Chang & Zuo 2025). Consequently, recent studies increasingly emphasize integrated socio-ecological perspectives, focusing not only on the ecological risks posed by *S. alterniflora*, but also on its dual roles in coastal protection, sediment stabilization, and carbon sequestration under changing environmental conditions.

In summary, the evolution of *S. alterniflora* research demonstrates a clear progression from species-centered ecological investigations toward integrated assessments of ecosystem functioning, climate interactions, and management trade-offs. This transition reflects the broader transformation of coastal wetland research in the Anthropocene, where invasive species are increasingly evaluated within coupled ecological, biogeochemical, and socio-environmental frameworks rather than solely as isolated biological disturbances.

4 Ecological consequences, management trade-offs, and future pathways

The bibliometric and thematic analyses presented above demonstrate that *S. alterniflora* research has progressively shifted from species-centered invasion studies toward broader investigations involving ecosystem functioning, climate interactions, and adaptive management. Building upon these thematic trends, the following sections further synthesize the ecological mechanisms, biogeochemical consequences, climate feedbacks, and management implications associated with *S. alterniflora* invasion across different coastal wetland systems.

4.1 Context-dependent ecological impacts across coastal wetlands

4.1.1 Salt marsh systems

Salt marshes represent the most extensively studied ecosystems affected by *S. alterniflora* invasion. In these environments, *S. alterniflora* often exhibits strong competitive advantages due to its high primary productivity, rapid clonal expansion, broad salinity tolerance, and efficient sediment-trapping capacity (Fig. 8) (S.-H. Li et al. 2025a, b). Numerous studies have demonstrated that invasion frequently results in the displacement of native salt-marsh vegetation, including species such as *Phragmites australis*, *Suaeda salsa*, and *Scirpus mariqueter*, thereby substantially altering plant community composition and ecosystem structure (Q.-W. Li et al. 2025a, b; Zhao et al. 2025).

The replacement of native vegetation by dense *S. alterniflora* monocultures can significantly modify marsh hydrology, sediment accretion processes, nutrient cycling, and habitat heterogeneity. In many estuarine marshes, invasion enhances sediment deposition and vertical accretion, which may increase marsh stability and resistance to erosion under moderate sea-level rise scenarios (Gong et al. 2021). Simultaneously, the extensive belowground

root system of *S. alterniflora* can improve sediment consolidation and organic matter accumulation, thereby influencing both geomorphological development and biogeochemical functioning (K.-H. Zhu et al. 2025a, b, c).

However, these ecosystem engineering effects are often accompanied by considerable ecological costs. Reductions in native plant diversity frequently cascade to higher trophic levels, including macrobenthic communities, fish assemblages, and migratory bird populations (Han et al. 2022; Zhang et al. 2020). In several coastal regions of China, invasion-induced habitat homogenization has been linked to declines in shorebird feeding habitats, particularly in open intertidal areas traditionally dominated by native halophytic vegetation (Y. Zhu et al. 2025a, b, c). Additionally, alterations in tidal creek morphology and sediment dynamics may reduce habitat complexity and modify ecological connectivity across marsh landscapes (Dong et al. 2024).

Importantly, the ecological impacts observed in salt marshes vary substantially among invasion stages. Early-stage invasions may temporarily enhance sediment retention and primary productivity without immediately reducing biodiversity, whereas long-term, high-density invasions often generate stronger competitive exclusion

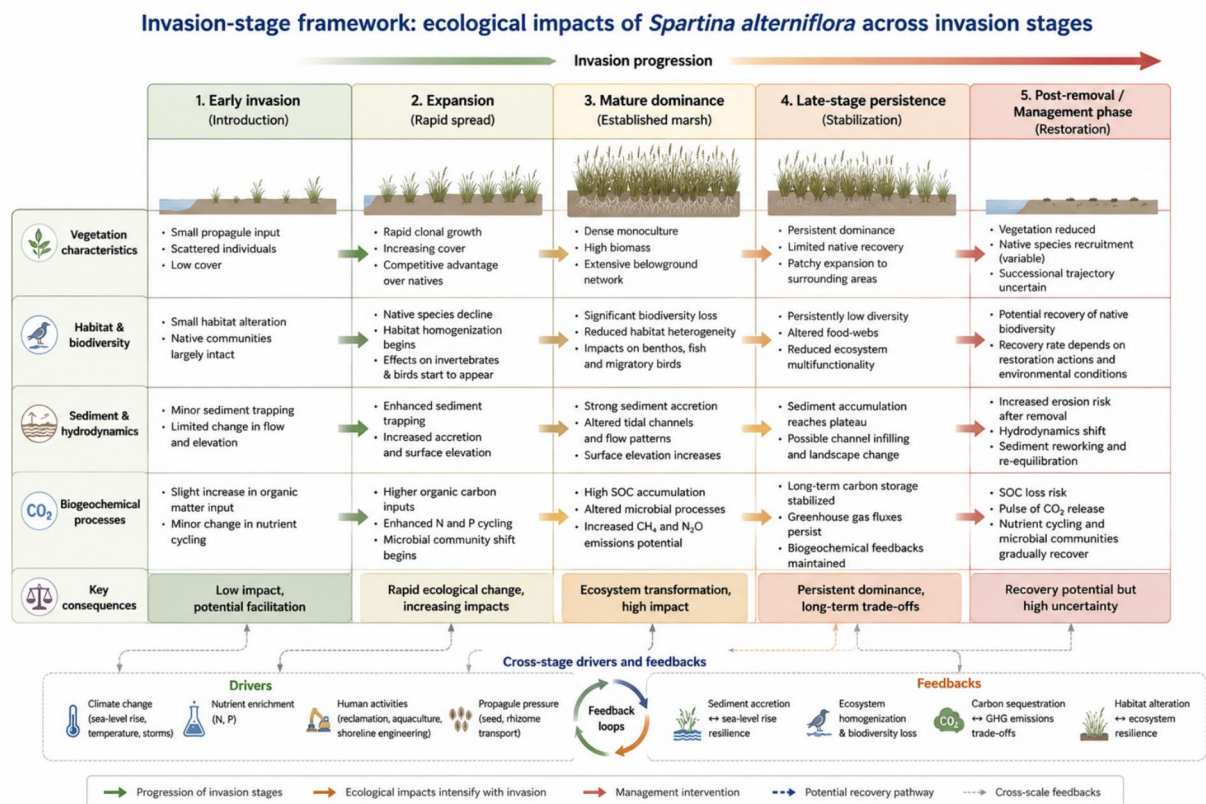


Fig. 8 Invasion-stage framework of *S. alterniflora*

effects and greater ecosystem homogenization. This temporal heterogeneity partly explains the contrasting conclusions reported among previous studies regarding the ecological consequences of *S. alterniflora* in salt marsh systems.

4.1.2 Mangrove ecosystems

The ecological interactions between *S. alterniflora* and mangrove ecosystems have attracted increasing attention in subtropical and tropical coastal regions, particularly in China, where large-scale mangrove restoration programs overlap spatially with *S. alterniflora* expansion. Compared with salt marshes, invasion dynamics in mangrove wetlands are more complex because of strong interactions among tidal inundation, canopy shading, sediment characteristics, and interspecific competition (Dong et al. 2024; Long et al. 2025).

In open mudflats adjacent to mangroves, *S. alterniflora* may initially facilitate sediment accretion and shoreline stabilization, potentially creating conditions favorable for subsequent mangrove establishment. This “facilitation effect” has been reported in some restoration-oriented studies, where moderate sediment accumulation promoted mangrove seedling recruitment and reduced wave exposure (Gu & Wu 2023). Nevertheless, under many circumstances, rapid *S. alterniflora* expansion competes directly with mangrove seedlings for space, nutrients, and light, thereby inhibiting mangrove regeneration and altering successional trajectories.

Dense *S. alterniflora* stands can also substantially modify sediment biogeochemistry and belowground microbial processes within mangrove-associated wetlands. Changes in redox conditions, organic matter accumulation, sulfur cycling, and microbial community composition have been widely documented following invasion (Mo et al. 2025). Such alterations may indirectly influence mangrove root development, nutrient availability, and ecosystem metabolism (Wang et al. 2014). Furthermore, invasion frequently modifies hydrological connectivity between tidal flats and mangrove forests, potentially affecting sediment transport pathways and estuarine nutrient exchange (Dong et al. 2024).

Recent studies have increasingly emphasized the importance of invasion stage and environmental gradients in determining invasion outcomes within mangrove ecosystems. For example, *S. alterniflora* tends to dominate lower elevation zones characterized by frequent tidal inundation, whereas mature mangrove forests with closed canopies often exhibit stronger resistance to invasion due to light limitation and competitive exclusion (Zhang et al. 2018). These findings suggest that ecological impacts cannot be generalized across all mangrove wetlands and must instead be interpreted within specific geomorphological and restoration contexts.

4.1.3 Tidal flats and estuarine wetlands

Tidal flats and estuarine wetlands are among the most vulnerable ecosystems affected by *S. alterniflora* invasion because of their dynamic hydrological regimes and ecological importance for migratory birds, benthic organisms, and nutrient exchange (Gong et al. 2024). In these systems, *S. alterniflora* often transforms unvegetated or sparsely vegetated intertidal flats into densely vegetated marsh habitats, thereby profoundly altering sedimentary environments and ecosystem processes (R. Wang et al. 2025a, b, c, d, e).

One of the most significant ecological consequences of invasion in tidal flats is the rapid shift from open mud-flat habitats to structurally simplified vegetated landscapes. Although vegetation establishment may enhance sediment stabilization and reduce shoreline erosion, it frequently results in the loss of critical feeding grounds for migratory shorebirds that depend on open intertidal habitats (Jackson et al. 2021). This issue has become particularly important along the East Asian–Australasian Flyway, where extensive *S. alterniflora* expansion has contributed to habitat degradation in internationally important migratory bird stopover sites (Gan et al. 2009).

Invasion also strongly influences benthic community composition and trophic interactions within estuarine ecosystems. Alterations in sediment texture, organic matter content, and oxygen availability can significantly affect macrobenthic diversity, meiofaunal assemblages, and microbial activity (W. Hu et al. 2026a, b; X. Zhang et al. 2025a, b). Some studies have reported increased abundance of opportunistic benthic species associated with organic matter enrichment, whereas others observed declines in native benthic biodiversity due to habitat modification and reduced environmental heterogeneity (Zhao et al. 2018; Zhu et al. 2019). These inconsistencies likely reflect differences in invasion duration, tidal dynamics, and regional environmental conditions.

At broader spatial scales, *S. alterniflora* invasion can alter estuarine geomorphology and hydrodynamic processes by modifying sediment transport, channel morphology, and tidal exchange (Ge et al. 2013). Such changes may influence nutrient retention, freshwater–saltwater interactions, and long-term coastal landscape evolution. Under accelerating sea-level rise and increasing anthropogenic disturbance, these hydrological and geomorphological feedbacks are expected to become increasingly important components of estuarine ecosystem resilience and coastal adaptation.

In essence, the ecological impacts of *S. alterniflora* across coastal wetlands are highly context-dependent and cannot be adequately characterized using a single generalized framework. The species simultaneously provides ecosystem functions such as sediment stabilization, shoreline

protection, and carbon accumulation while also generating substantial ecological risks including biodiversity loss, habitat homogenization, and altered trophic dynamics. These dual and often conflicting effects underscore the necessity of ecosystem-specific assessment and adaptive management strategies when evaluating the long-term role of *S. alterniflora* in coastal wetlands under global environmental change.

4.2 Biogeochemical consequences and blue carbon uncertainties

4.2.1 Carbon sequestration mechanisms

The high carbon sequestration potential associated with *S. alterniflora* invasion is primarily attributed to its exceptional primary productivity, extensive belowground biomass, and strong sediment-trapping capacity. Compared with many native coastal marsh species, *S. alterniflora* typically exhibits rapid growth rates and dense root-rhizome systems that contribute substantial organic carbon inputs to wetland soils (Zhang et al. 2026). The accumulation of belowground biomass is particularly important because root-derived carbon is generally more resistant to decomposition and can remain stored in sediments over relatively long timescales (Tang et al. 2026).

In addition to direct biomass production, *S. alterniflora* significantly influences sedimentary carbon burial through its capacity to modify hydrodynamic conditions and enhance sediment accretion (Gao et al. 2020). Dense vegetation can reduce current velocity and wave energy, thereby promoting the deposition of suspended sediments and associated organic matter (Zhao et al. 2023). Increased vertical sediment accretion may further improve marsh resilience to sea-level rise while simultaneously enhancing long-term carbon storage potential. Several studies conducted in estuarine wetlands of China and North America have reported substantially higher soil organic carbon (SOC) contents and carbon accumulation rates in *S. alterniflora*-invaded marshes compared with adjacent unvegetated tidal flats or native vegetation communities (Song et al. 2023).

The invasion of *S. alterniflora* also affects carbon cycling indirectly through alterations in microbial processes and nutrient dynamics. Changes in rhizosphere oxygen conditions, dissolved organic carbon availability, and microbial community composition can strongly influence rates of organic matter decomposition and carbon stabilization (Liu et al. 2026). In some cases, invasion has been associated with increased microbial biomass and enhanced carbon turnover, whereas in others, reduced decomposition rates under anoxic sediment conditions may facilitate long-term carbon preservation (Song et al. 2025; X.-Y. Wang et al. 2024a, b, c, d, e).

Nitrogen enrichment and plant–microbe interactions further contribute to the biogeochemical consequences of invasion (Jin et al. 2017). Enhanced nitrogen availability may stimulate plant productivity and carbon inputs, while microbial functional shifts can alter the balance between carbon mineralization and stabilization (Dai et al. 2024; Long et al. 2025). Consequently, the carbon sequestration capacity of *S. alterniflora* wetlands emerges from the combined effects of vegetation growth, sediment dynamics, and belowground biogeochemical regulation rather than from biomass accumulation alone.

Recent studies have increasingly emphasized that carbon sequestration responses vary considerably among wetland types, climatic regions, and invasion stages. For example, early invasion stages may rapidly increase sediment accumulation and carbon burial, whereas mature invasions may eventually reach saturation or generate altered decomposition dynamics (Gu et al. 2025; Zhang et al. 2026). Similarly, the magnitude of carbon accumulation differs substantially between salt marshes, mangrove-associated wetlands, and estuarine tidal flats because of differences in hydrology, sediment supply, and vegetation structure (Hou et al. 2026). These findings highlight the necessity of evaluating blue carbon potential within ecosystem-specific environmental contexts.

4.2.2 Uncertainties in blue carbon accounting

Although *S. alterniflora* invasion is frequently associated with enhanced carbon storage, considerable uncertainty remains regarding its net climate mitigation potential. One major limitation in current assessments is the predominant focus on soil organic carbon accumulation, while other important carbon pathways and greenhouse gas fluxes are often insufficiently quantified (Liu et al. 2017).

Methane (CH₄) emissions represent one of the most significant uncertainties in evaluating the net climatic effects of *S. alterniflora* invasion (Gong et al. 2025). Increased organic matter accumulation and altered redox conditions within waterlogged sediments may stimulate methanogenic microbial activity, potentially offsetting part of the climate benefits associated with carbon sequestration (Dong et al. 2023). Several studies have reported elevated methane emissions in invaded wetlands, although the magnitude and direction of these responses vary substantially among sites and environmental conditions (Wilburn et al. 2024; X. Zhu et al. 2025a, b, c). Nitrous oxide (N₂O) emissions may also increase under nitrogen-enriched conditions, further complicating assessments of net greenhouse gas balance (Chen et al. 2026).

Another important source of uncertainty involves the “priming effect,” whereby fresh organic carbon inputs from highly productive *S. alterniflora* vegetation stimulate microbial decomposition of previously stored soil carbon. In such cases, apparent increases in carbon inputs may partially accelerate the mineralization of older carbon pools, thereby reducing the net efficiency of long-term carbon sequestration (Li et al. 2026; L. Wang et al. 2025a, b, c, d, e). However, the strength and persistence of priming effects remain poorly constrained across different coastal wetland environments.

Current blue carbon assessments also frequently overlook the potential role of soil inorganic carbon (SIC), particularly in estuarine and saline environments where carbonate dynamics may contribute substantially to total sediment carbon storage. Most existing studies focus exclusively on SOC measurements, potentially underestimating or misrepresenting the overall carbon budget of invaded wetlands (Xie & Li 2024). Moreover, variations in sediment mineral composition, salinity, and alkalinity can strongly influence carbonate precipitation and dissolution processes, introducing additional spatial variability into carbon accounting (Zhang et al. 2021).

Methodological inconsistencies further contribute to uncertainty across studies. Differences in sampling depth, carbon density calculation methods, temporal monitoring duration, and greenhouse gas measurement techniques often limit direct comparison among regions (Jin et al. 2016; D. Wang et al. 2025a, b, c, d, e). In many cases, short-term observations are used to infer long-term sequestration potential, despite substantial temporal variability associated with seasonal dynamics, disturbance events, and invasion progression. Furthermore, many studies rely on site-specific measurements that may not adequately capture broader regional heterogeneity in hydrology, sediment supply, or climatic conditions (Ding et al. 2023).

The ecological trade-offs associated with blue carbon enhancement also remain insufficiently resolved. While *S. alterniflora* invasion may increase sediment carbon storage and shoreline stabilization, these benefits are frequently accompanied by biodiversity loss, habitat homogenization, and altered trophic interactions (D. Wang et al. 2025a, b, c, d, e). Consequently, evaluating the ecological value of *S. alterniflora* solely from a carbon sequestration perspective may oversimplify the broader socio-ecological implications of invasion. A synthesis of quantitative evidence on blue carbon dynamics associated with *S. alterniflora* invasion is provided in Table 2, highlighting strong spatial–temporal variability and contrasting climatic effects across coastal wetland types.

All in all, existing evidence suggests that *S. alterniflora* possesses substantial potential to influence coastal carbon cycling and blue carbon accumulation, but its net climatic and ecological consequences remain highly context-dependent. Future research should prioritize integrated carbon accounting frameworks that simultaneously quantify soil organic and inorganic carbon, greenhouse gas fluxes, microbial processes, and long-term ecosystem dynamics across multiple spatial and temporal scales. Such approaches are essential for accurately assessing the role of *S. alterniflora* within coastal climate mitigation and adaptive management strategies under accelerating global environmental change.

4.3 Climate change interactions and invasion feedbacks

Climate change and *Spartina alterniflora* invasion are increasingly recognized as interconnected processes that jointly influence the structure, functioning, and resilience of coastal wetlands. Rather than acting independently, climatic drivers and biological invasion frequently interact through multiple ecological feedback mechanisms, thereby reshaping coastal ecosystem dynamics across spatial and temporal scales (Cui et al. 2025; Liu et al.

Table 2 Quantitative evidence of blue carbon effects associated with *S. alterniflora* invasion across coastal wetland ecosystems

Ecosystem type	Indicator	Reported effect	Direction	Representative studies
Salt marsh	Soil organic carbon (SOC) accumulation	+ ~ 10% to + ~ 60% (with higher site-specific values reported)	Increase	(Wang et al. 2021; Yang et al. 2017)
Salt marsh	Soil inorganic carbon (SIC)	High variability across sites	Uncertain	(Bu et al. 2018; Yang & Yang 2020)
Mangrove	Soil organic carbon (SOC) accumulation	Context dependent	Increase or decrease	(Liu et al. 2023; F. Wang et al. 2024d, e, a, b, c)
Tidal flat	Sediment carbon accumulation	~ 0.5 – ~ 3.5 Mg C ha ⁻¹ yr ⁻¹	Increase	(Chen et al. 2023)
Coastal wetlands	Methane (CH ₄) emissions	+ ~ 20% to + ~ 300% relative increase	Increase (warming effect)	(An et al. 2024; Weston et al. 2014; Yao et al. 2023)
Coastal wetlands	Nitrous oxide (N ₂ O) emissions	Variable across studies	Uncertain	(Xu et al. 2017; Yao et al. 2023)
Coastal wetlands	Net greenhouse gas balance (CO ₂ -eq)	Variable (net sink to net source depending on conditions)	Highly uncertain	(Pastore et al. 2017)

* Direction refers to net change relative to native vegetation or unvegetated reference systems, and may vary depending on hydrological and climatic conditions

2025). These interactions are particularly important in estuarine and intertidal environments, where hydrological conditions, sediment processes, and vegetation responses are highly sensitive to global environmental change.

Sea-level rise is widely considered one of the most significant climatic factors influencing the distribution and expansion of *S. alterniflora* (J. Wang et al. 2025a, b, c, d, e). Owing to its strong tolerance to tidal inundation and rapid sediment-trapping capacity, *S. alterniflora* is often capable of colonizing low-elevation intertidal zones more effectively than many native coastal plant species (J. Yang et al. 2025a, b). Enhanced sediment accretion associated with dense vegetation can increase marsh surface elevation and partially offset moderate sea-level rise, thereby improving short-term wetland stability and resistance to erosion. In this context, some studies have suggested that *S. alterniflora* may contribute to coastal protection and marsh persistence under accelerating sea-level rise scenarios (Williams & Johnson 2021; M. Zhang et al. 2023a, b).

However, the ecological consequences of this adaptive capacity are highly complex. The expansion of *S. alterniflora* under rising sea levels may accelerate the conversion of open tidal flats into vegetated marshes, leading to the loss of critical habitats for migratory shorebirds and other intertidal organisms (Liu et al. 2025). In many estuarine wetlands, sea-level rise and invasion may therefore act synergistically to intensify habitat homogenization and reduce biodiversity. Moreover, increasing tidal inundation and shoreline retreat may facilitate inland migration of *S. alterniflora*, particularly in areas where coastal squeeze and anthropogenic barriers restrict the natural landward movement of native wetlands (Liu et al. 2025).

Temperature increases associated with global warming may further enhance the invasion potential of *S. alterniflora* by extending its suitable climatic range and lengthening the growing season (Fan et al. 2025). Predictive distribution models increasingly indicate potential poleward expansion under future warming scenarios, particularly in temperate coastal regions currently constrained by low winter temperatures (Acosta-Rodriguez et al. 2026). Elevated temperatures may also alter plant productivity, microbial activity, and nutrient cycling, thereby influencing ecosystem carbon dynamics and greenhouse gas emissions (Y. Wu et al. 2026a, b). Nevertheless, regional responses remain highly variable because of differences in hydrology, sediment supply, salinity stress, and disturbance regimes.

Extreme climatic events, including storm surges, typhoons, heatwaves, and droughts, also play important roles in shaping invasion dynamics and ecosystem resilience. In some coastal wetlands, storm-driven sediment

deposition may create newly suitable habitats for *S. alterniflora* colonization, whereas prolonged drought or hypersaline conditions may suppress vegetation growth and reduce competitive advantages (Agrelus et al. 2018; Sengupta et al. 2019; Wan et al. 2009). Similarly, disturbances associated with tropical cyclones can either facilitate invasion through habitat fragmentation or reduce biomass through physical damage, depending on local environmental conditions and invasion stage.

Importantly, *S. alterniflora* not only responds to climate change but also actively modifies ecosystem responses to climatic stressors. Through its influence on sediment accretion, hydrodynamics, carbon sequestration, and nutrient cycling, invasion can alter the adaptive capacity and resilience of coastal wetlands under changing environmental conditions. For example, increased sediment stabilization may enhance shoreline resistance to erosion, while altered vegetation structure and hydrological connectivity may simultaneously reduce habitat diversity and ecosystem flexibility (Gong et al. 2026; Ning et al. 2023). These bidirectional feedbacks indicate that *S. alterniflora* should be viewed not merely as a passive responder to climate change, but as an active ecological driver capable of reshaping coastal adaptation trajectories.

Despite increasing research attention, substantial uncertainties remain regarding the long-term interactions between climate change and *S. alterniflora* invasion. Many existing studies are limited by short-term observations, localized spatial scales, and insufficient integration of ecological, hydrological, and climatic processes. Future research should therefore prioritize coupled modeling approaches, long-term monitoring, and multi-scale assessments that explicitly incorporate invasion dynamics into broader coastal climate adaptation frameworks.

4.4 Multi-scale ecological trade-offs and management dilemmas

The management of *S. alterniflora* presents a complex ecological dilemma because the species simultaneously provides important ecosystem functions while generating substantial ecological risks. Traditional invasion management frameworks often emphasize complete eradication of invasive species (Gong et al. 2026); however, the ecological role of *S. alterniflora* in coastal wetlands is increasingly recognized as multidimensional and context-dependent (Li et al. 2024). Consequently, management decisions must balance competing ecological, climatic, economic, and social objectives across multiple spatial and temporal scales.

One of the most significant trade-offs associated with *S. alterniflora* invasion involves the balance between coastal protection and biodiversity conservation. Dense *S. alterniflora* vegetation can effectively reduce wave

energy, stabilize shorelines, and enhance sediment deposition, thereby contributing to erosion control and wetland accretion (FuChen & Fang 2007; Mavrodi et al. 2018). These functions may become increasingly valuable under accelerating sea-level rise and intensifying coastal disturbances. In some regions, particularly highly engineered coastlines vulnerable to erosion, the sediment-trapping capacity of *S. alterniflora* has been regarded as beneficial for maintaining shoreline stability and supporting land formation processes.

At the same time, these geomorphological benefits are frequently accompanied by substantial ecological costs. The conversion of open tidal flats and heterogeneous marsh landscapes into dense monocultures often leads to reductions in native plant diversity, habitat simplification, and altered trophic interactions (Zhu et al. 2026). Migratory shorebirds, benthic fauna, and estuarine food webs are particularly vulnerable to habitat modification associated with invasion (Lyu et al. 2023; Zhu et al. 2026). Consequently, management priorities may differ considerably depending on whether the primary objective is biodiversity protection, shoreline stabilization, carbon sequestration, or ecosystem restoration.

Additional trade-offs emerge in the context of blue carbon and climate mitigation. Although *S. alterniflora* invasion may enhance sediment carbon storage and marsh accretion, these potential climate benefits remain entangled with uncertainties regarding methane emissions, biodiversity loss, and long-term ecosystem sustainability. The increasing emphasis on blue carbon has therefore generated debate over whether carbon sequestration benefits should influence invasion management strategies. Some studies argue that carbon accumulation and coastal resilience functions should be incorporated into management assessments, whereas others emphasize that prioritizing carbon storage alone may underestimate broader ecological degradation and conservation risks (Wei et al. 2024; Wu et al. 2025). As summarized in Fig. 9, the ecological outcomes of *Spartina alterniflora* invasion emerge from dynamic interactions among biodiversity conservation, shoreline stabilization, carbon sequestration, climate adaptation, and management interventions. Management effectiveness is further complicated by strong regional heterogeneity in invasion dynamics and ecosystem responses. In some estuarine wetlands, partial retention or controlled utilization of *S. alterniflora* may provide temporary ecological or engineering benefits, while in ecologically sensitive areas such as migratory bird habitats and protected tidal flats, rapid eradication may be necessary to preserve ecosystem integrity. These contrasting outcomes suggest that universal management strategies are unlikely to be effective across all coastal wetland types.

Large-scale eradication programs also face significant technical, economic, and ecological challenges. Mechanical removal, herbicide application, flooding regulation, and ecological replacement have all been employed to control *S. alterniflora* invasion, yet each approach presents limitations related to cost, environmental disturbance, long-term effectiveness, and secondary ecological impacts (Adams et al. 2016; Chang & Zuo 2025). In some cases, rapid removal may destabilize sediments, increase erosion risk, or facilitate secondary invasions by other opportunistic species (Y. Hu et al. 2026a, b). Post-removal ecosystem recovery also remains highly uncertain, particularly in heavily altered estuarine environments where native vegetation recruitment and hydrological restoration are constrained (J. Wang et al. 2024a, b, c, d, e).

Recent management frameworks increasingly advocate adaptive and ecosystem-based approaches that integrate ecological restoration, long-term monitoring, and regional environmental objectives. In China, for example, large-scale *S. alterniflora* control initiatives implemented since 2022 have accelerated research on integrated management strategies combining invasion control with wetland restoration and biodiversity conservation. Simultaneously, advances in remote sensing, UAVs, machine learning, and spatial ecological modeling have improved the capacity to monitor invasion dynamics and evaluate management outcomes across large coastal regions (Chang & Zuo 2025; Zhang et al. 2026).

In brief, the ecological and management implications of *S. alterniflora* cannot be adequately assessed through binary classifications of “harmful” or “beneficial.” Instead, invasion outcomes reflect complex trade-offs among ecosystem functioning, biodiversity conservation, climate adaptation, and socio-economic priorities. Future management strategies should therefore emphasize context-dependent decision-making frameworks that account for regional environmental conditions, invasion stage, ecological sensitivity, and long-term coastal sustainability under global environmental change.

4.5 Emerging technologies and future research priorities

Recent advances in observation technologies, ecological modeling, and interdisciplinary analytical approaches are substantially transforming *S. alterniflora* research. Compared with earlier studies that primarily relied on localized field surveys and descriptive ecological observations, contemporary research increasingly integrates multi-source remote sensing, molecular ecology, artificial intelligence, and coupled socio-ecological modeling to investigate invasion dynamics and ecosystem responses across broader spatial and temporal scales (Chang & Zuo 2025; Kong et al. 2025). These emerging technologies provide new

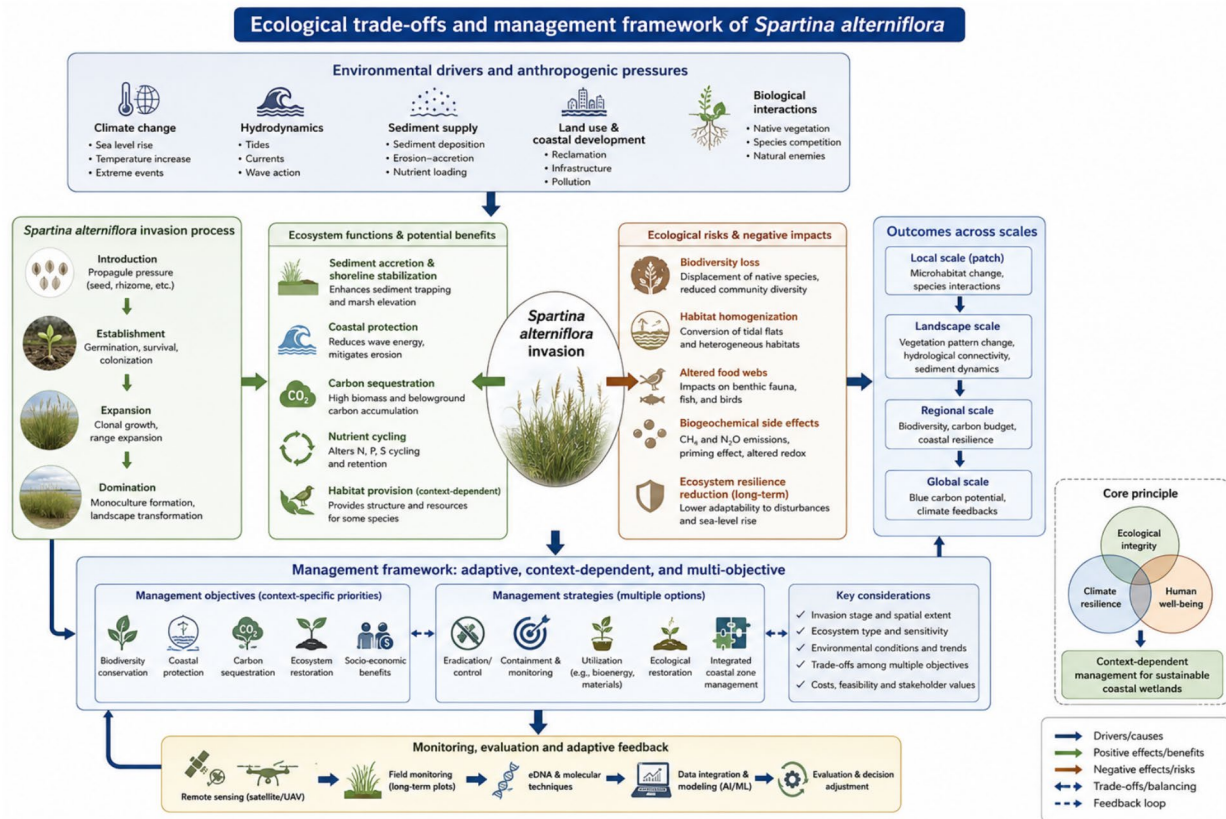


Fig. 9 Ecological trade-offs and management framework of *S. alterniflora*

opportunities to address long-standing uncertainties regarding invasion mechanisms, climate feedbacks, ecological trade-offs, and adaptive management.

Remote sensing technologies have become indispensable tools for large-scale monitoring of *S. alterniflora* distribution and invasion dynamics. Satellite platforms with high spatial and temporal resolution, including Landsat, Sentinel, and Gaofen series imagery, have significantly improved the capacity to detect vegetation expansion, quantify landscape changes, and assess habitat transformation in coastal wetlands (Chen et al. 2020; Min et al. 2025; Zhou et al. 2024). In particular, time-series remote sensing analyses enable long-term reconstruction of invasion trajectories and facilitate the identification of invasion hotspots under changing environmental conditions.

The rapid development of UAVs and low-altitude hyperspectral imaging has further enhanced monitoring precision at local and regional scales. UAV-based observations can provide centimeter-level spatial resolution, enabling detailed assessments of vegetation structure, biomass distribution, canopy characteristics, and restoration outcomes (Chang & Zuo 2025). Combined with machine learning and deep learning algorithms, these technologies increasingly support automated

classification and early detection of invasion fronts, thereby improving the efficiency of ecological monitoring and management evaluation.

Artificial intelligence and spatial ecological modeling are also becoming increasingly important in predicting future invasion risks and ecosystem responses under climate change scenarios. Species distribution models, ecological niche models, and coupled hydrodynamic-ecological simulations have been widely applied to evaluate potential expansion pathways, habitat suitability shifts, and interactions between invasion processes and sea-level rise (N. Zhang et al. 2025a, b). The integration of climate projections, sediment dynamics, and vegetation competition into predictive models provides valuable tools for scenario analysis and adaptive coastal management planning.

At the same time, molecular and microbial ecological approaches are substantially advancing mechanistic understanding of invasion processes. High-throughput sequencing, metagenomics, and environmental DNA (eDNA) technologies increasingly reveal how *S. alterniflora* invasion alters microbial community composition, functional gene expression, and belowground biogeochemical pathways (Chen et al. 2026; Sui et al. 2025; Sun

et al. 2025). These approaches provide critical insights into plant–microbe interactions, nutrient cycling, sulfur and nitrogen transformations, and greenhouse gas production within invaded wetlands. Recent studies also suggest that microbial feedbacks may play important roles in determining invasion success, ecosystem resilience, and carbon sequestration potential (Y. Wang et al. 2025a, b, c, d, e; G. Zhang et al. 2024a, b).

Despite the rapid development of emerging technologies and interdisciplinary analytical approaches, several critical knowledge gaps remain unresolved in current *S. alterniflora* research. One of the major limitations lies in the insufficient availability of long-term ecosystem observations across diverse coastal regions. Most existing studies are still based on short-term investigations or localized case studies, which may not adequately capture the long-term ecological trajectories associated with invasion progression, climate variability, and post-restoration dynamics. Expanding integrated long-term monitoring networks across multiple wetland types and climatic gradients will therefore be essential for improving understanding of ecosystem responses under accelerating global environmental change.

At the same time, future research should increasingly move beyond isolated disciplinary perspectives toward more integrative and cross-scale analytical frameworks. The ecological consequences of *S. alterniflora* invasion are closely coupled with hydrodynamic processes, geomorphological evolution, climate adaptation, ecosystem service provision, and human coastal development. However, these interconnected processes are still frequently examined independently, limiting comprehensive understanding of how biological invasion influences coastal resilience, biodiversity conservation, carbon cycling, and socio-economic sustainability simultaneously. Greater integration among ecological, biogeochemical, hydrological, and socio-economic approaches will therefore be necessary to address the complexity of coastal socio-ecological systems.

Considerable uncertainties also remain regarding the long-term effectiveness and ecological consequences of invasion management and restoration practices. Although large-scale control programs have been widely implemented in many coastal regions, their long-term ecological outcomes remain insufficiently evaluated. Future studies should place greater emphasis on adaptive management experiments, ecosystem recovery trajectories, and post-removal ecological stability, particularly with respect to sediment dynamics, native vegetation recovery, secondary invasions, and long-term biodiversity responses.

Similarly, current blue carbon assessments remain constrained by methodological inconsistencies and

incomplete carbon accounting frameworks. Future studies should increasingly integrate soil organic carbon, soil inorganic carbon, greenhouse gas emissions, microbial carbon turnover, and sediment accretion processes within unified analytical frameworks. Such integrated assessments are necessary for accurately evaluating the net climatic effects of *S. alterniflora* invasion across different environmental contexts and temporal scales.

More broadly, future *S. alterniflora* research should increasingly adopt an Anthropocene-oriented perspective that recognizes the species not only as an invasive ecological stressor, but also as a potential component of coastal adaptation processes under climate change. Rather than relying on binary classifications of ecological harm versus ecological benefit, future studies should focus more explicitly on the dynamic trade-offs among ecosystem resilience, shoreline protection, biodiversity conservation, carbon sequestration, and climate mitigation. Advancing toward integrated, context-dependent, and adaptive research frameworks will be critical not only for improving understanding of *Spartina alterniflora* invasion, but also for informing sustainable management of coastal socio-ecological systems under accelerating Anthropocene change.

5 Conclusions

This study synthesized the current knowledge structure, research evolution, ecological impacts, and management challenges associated with *S. alterniflora* invasion through a quantitative systematic review integrating bibliometric analysis and thematic synthesis. The results demonstrate that *S. alterniflora* research has undergone a substantial transformation over the past four decades, shifting from species-centered invasion ecology toward increasingly interdisciplinary investigations involving biogeochemistry, climate change, ecosystem resilience, and adaptive coastal management.

The review highlights that the ecological consequences of *S. alterniflora* invasion are highly context-dependent and vary considerably across coastal wetland types, climatic regions, and invasion stages. In salt marshes, mangrove ecosystems, and tidal flats, *S. alterniflora* simultaneously functions as an ecosystem engineer capable of enhancing sediment accretion, shoreline stabilization, and carbon accumulation, while also generating substantial ecological risks including biodiversity decline, habitat homogenization, and altered trophic interactions. These dual and often conflicting effects underscore the necessity of evaluating invasion outcomes within ecosystem-specific environmental and management contexts.

This study also demonstrates that the growing emphasis on blue carbon and climate adaptation has

substantially reshaped contemporary *S. alterniflora* research. Although invasion may increase carbon storage and wetland accretion, significant uncertainties remain regarding net climatic benefits because of methane emissions, priming effects, spatial heterogeneity, and incomplete carbon accounting approaches. Consequently, future blue carbon assessments should integrate soil organic carbon, soil inorganic carbon, greenhouse gas fluxes, microbial processes, and long-term ecosystem dynamics within unified analytical frameworks.

The interactions between climate change and *S. alterniflora* invasion further reveal complex bidirectional feedbacks between biological invasion and coastal ecosystem resilience. Sea-level rise, warming temperatures, and extreme climatic events may facilitate invasion expansion, while invasion itself can modify sediment dynamics, hydrological processes, and coastal adaptive capacity. These findings suggest that *S. alterniflora* should be understood not merely as a passive responder to climate change, but as an active ecological driver influencing future coastal landscape evolution.

This review emphasizes that effective management of *S. alterniflora* requires adaptive, ecosystem-based, and region-specific strategies rather than universally applied eradication approaches. Management decisions should balance biodiversity conservation, shoreline protection, carbon sequestration, ecological restoration, and socio-economic priorities under changing environmental conditions. Emerging technologies, including remote sensing, UAV monitoring, artificial intelligence, and coupled ecological modeling, are expected to play increasingly important roles in supporting long-term monitoring and adaptive management. *S. alterniflora* represents a representative example of the complex ecological trade-offs that characterize coastal wetlands in the Anthropocene. Future research should therefore move beyond binary classifications of invasive species impacts and instead focus on integrated socio-ecological frameworks capable of addressing the interconnected challenges of climate change, ecosystem resilience, biodiversity conservation, and sustainable coastal management.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44218-026-00142-z>.

Supplementary Material 1.

Acknowledgements

Not applicable.

Authors' contributions

H. C. and P. Z. designed and conceptualised the review. H. C. organised the manuscript. H. C. oversaw the design and production of the major figures.

H. C., P. Z., S.-S. C., T.-X. Z., J.-H. L., J.-H. T., and P. Q. wrote the manuscript. All authors reviewed and approved the final version.

Funding

This work was supported by the Marine Science & Technology Innovation Project of Jiangsu Department of Natural Resources with Grant No. JSZ-RHYKJ202213; the National Key Research and Development Program of China under Grant No. 2017YFC0506506.

Data availability

All sources of information are included in this review article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Received: 6 January 2026 Revised: 17 June 2026 Accepted: 22 June 2026
Published online: 15 July 2026

References

- Acosta-Rodriguez VN, Cole R, Doomstorm D, Wittingham S. S., Johnson DS (2026) Northern range expansion of the Atlantic mangrove fiddler crab *Leptuca thayeri* Rathbun, 1900 (Decapoda: Brachyura: Ocypodidae). *J Crustacean Biol* 46(1):ruaf072. <https://doi.org/10.1093/jcbiol/ruaf072>
- Adams J, van Wyk E, Riddin T (2016) First record of *Spartina alterniflora* in southern Africa indicates adaptive potential of this saline grass [Article]. *Biol Invasions* 18(8):2153–2158. <https://doi.org/10.1007/s10530-015-0957-5>
- Agrelus T, Dudycha JL, Morris JT (2018) Global DNA cytosine methylation variation in *Spartina alterniflora* at North Inlet, SC [Article]. *PLOS ONE* 13(9): e0203230. <https://doi.org/10.1371/journal.pone.0203230>
- An Z, Chen F, Zheng Y, Zhou J, Liu B, Qi L, Lin Z, Yao C, Wang B, Wang Y, Li X, Yin G, Dong H, Liang X, Liu M, Hou L (2024) Role of n-DAMO in Mitigating Methane Emissions from Intertidal Wetlands Is Regulated by Saltmarsh Vegetations [Article]. *Environ Sci Technol* 58(2):1152–1163. <https://doi.org/10.1021/acs.est.3c07882>
- Aria M, Cuccurullo C (2017) bibliometrix: An R-tool for comprehensive science mapping analysis. *J Inform* 11(4): 959–975. (Original work published Elsevier)
- Ayres DR, Garcia-Rossi D, Davis HG, Strong DR (1999) Extent and degree of hybridization between exotic (*Spartina alterniflora*) and native (*S. foliosa*) cordgrass (Poaceae) in California, USA determined by random amplified polymorphic DNA (RAPDs) [Article]. *Mol Ecol* 8(7):1179–1186. <https://doi.org/10.1046/j.1365-294x.1999.00679.x>
- Bu N, Yang X, Li G, Ma X, Song Y, Ma F, Li B, Fang C, Yan Z (2018) Effects of *Spartina alterniflora* invasion on soil carbon dynamics in wetlands of the Yangtze River estuary. *China Environ Sci* 38(7):2671–2679, Article 1000–6923(2018)38:7<2671:Hhmcrcq>2.0.Tx;2-y
- Chang H, Zuo P (2025) Review of research on *Spartina alterniflora* in coastal wetlands based on Unmanned Aerial Vehicle (UAV) [Review]. *Nat Remote Sens Bull* 29(5):1074–1088, Article 1007–4619(2025)29:5<1074:Bhshdh>2.0.Tx;2-s.

- Chang W, Cheng J, Allaire J, Sievert C, Schloerke B, Xie Y, Allen J, McPherson J, Dipert A, Borges B (2023) Shiny: Web Application Framework for R. R package version 1.8.0. <https://CRAN.R-project.org/package=shiny>.
- Chen Z, Guo L, Jin B, Wu J, Zheng G (2009) Effect of the exotic plant *Spartina alterniflora* on macrobenthos communities in salt marshes of the Yangtze River Estuary, China [Article]. *Estuarine Coast Shelf Sci* 82(2):265–272. <https://doi.org/10.1016/j.ecss.2009.01.014>
- Chen M, Ke Y, Bai J, Li P, Lyu M, Gong Z, Zhou D (2020) Monitoring early stage invasion of exotic *Spartina alterniflora* using deep-learning super-resolution techniques based on multisource high-resolution satellite imagery: A case study in the Yellow River Delta, China [Article]. *Int J Appl Earth Observ Geoinform* 92:102180. <https://doi.org/10.1016/j.jag.2020.102180>
- Chen Y, Zhang Z, Chen L, Zhang J, Liu B, Xia X, Wang X, Cai T (2023) Seasonal variation in coastal saltmarsh carbon stocks, south bank of Hangzhou Bay. *J Marine Sci* 41(1):55–67, Article 1001–909x(2023)41:1<55:Hwnab>2.0.Tx;2-g
- Chen S, Wu H, Xu W, Wang F, Yan R, Li S, Chen N (2026) Biogenic carbon input from *Spartina alterniflora* eradication governs nitrate removal pathways and N₂O emission in estuarine mangrove wetlands [Article]. *Geoderma* 468:117780. <https://doi.org/10.1016/j.geoderma.2026.117780>
- Cheng Z, Han L, Lv Z, Tian J, Gao J, Zhang L (2026) The Synergistic Effect of Hydrothermal and Deep Eutectic Solvent Pretreatment on the Resource Utilization of *Spartina alterniflora* Straw [Article]. *Chem Select* 11(13):e73121. <https://doi.org/10.1002/slct.73121>
- Chung C-H (2006) Forty years of ecological engineering with *Spartina* plantations in China [Article]. *Ecol Eng* 27(1):49–57. <https://doi.org/10.1016/j.ecoleng.2005.09.012>
- Chung C-H (1993) Thirty years of ecological engineering with *Spartina* plantations in China. *Ecol Eng* 2(3):261–289. [https://doi.org/10.1016/0925-8574\(93\)90019-C](https://doi.org/10.1016/0925-8574(93)90019-C)
- Cui L, Qiu J, Berger U, Cao M, Li W, Jiang J (2025) Comparing and quantifying the ecological niches of the saltmarsh grass *Spartina alterniflora* and major mangrove species in China [Article]. *Sci Rep* 15(1):23604. <https://doi.org/10.1038/s41598-025-07784-y>
- Dai Z, Zhang N, Wang F, Li Y, Peng J, Xiang T, Zhao X, Yang S, Cao W (2024) Loss of microbial functional diversity following *Spartina alterniflora* invasion reduces the potential of carbon sequestration and nitrogen removal in mangrove sediments from a gene perspective [Article]. *J Environ Manag* 365:121569. <https://doi.org/10.1016/j.jenvman.2024.121569>
- Diagne C, Leroy B, Vaissière A-C, Gozlan RE, Roiz D, Jarić I, Salles J-M, Bradshaw CJA, Courchamp F (2021) High and rising economic costs of biological invasions worldwide. *Nature* 592(7855):571–576. <https://doi.org/10.1038/s41586-021-03405-6>
- Ding X, Wang W, Wen J, Feng T, Penuelas J, Sardans J, Liang C, Agathokleous E, Wang C, Song Z, Li Q, Filley T. R, He H, Zhang X (2023) *Spartina alterniflora* invasion differentially alters microbial residues and their contribution to soil organic C in coastal marsh and mangrove wetlands [Article]. *Catena* 230:107246. <https://doi.org/10.1016/j.catena.2023.107246>
- Dong Y, Yuan J, Li J, Liu D, Qiu Y, Zhang X, Xiang J, Ding W (2023) Conversion of natural coastal wetlands to mariculture ponds dramatically decreased methane production by reducing substrate availability [Article]. *Agric Ecosyst Environ* 356:108646. <https://doi.org/10.1016/j.agee.2023.108646>
- Dong D, Gao Q, Huang H (2024) Mangroves Invaded by *Spartina alterniflora* Loisel: A Remote Sensing-Based Comparison for Two Protected Areas in China [Article]. *Forests* 15(10):1788. <https://doi.org/10.3390/f15101788>
- Dong D, Huang H, Gao Q (2024) Tracking the Dynamics of *Spartina alterniflora* with WorldView-2/3 and Sentinel-1/2 Imagery in Zhangjiang Estuary, China [Article]. *Water* 16(13):1780. <https://doi.org/10.3390/w16131780>
- Elton CS (1958) The ecology of invasions by animals and plants. Methuen
- Fan L, Cheng J, Krasilnikov P, Xu L, Wu Y, Xu J (2025) Effects of *Spartina alterniflora* invasion on carbon fluxes in coastal salt marsh ecosystems in China [Article]. *Estuarine Coast Shelf Sci* 327:109591. <https://doi.org/10.1016/j.ecss.2025.109591>
- FuChen SHI, Fang BAO (2007) Effects of salt and temperature stress on ecophysiological characteristics of exotic cordgrass, *Spartina alterniflora*. *Acta Ecologica Sinica* 27(7):2733–2741, Article 1000–0933(2007)27:7<2733:Yhwpx>2.0.Tx;2-n. [https://doi.org/10.1016/S1872-2032\(07\)60061-4](https://doi.org/10.1016/S1872-2032(07)60061-4)
- Gan X, Zhang K, Ma Z, Chen J, Li B (2006) The effect of invasions of the grass *Spartina alterniflora* on wintering birds on Chongming Island, Dongtan Reserve, China [Meeting Abstract]. *J Ornithol* 147(5):169–169
- Gan X, Cai Y, Choi C, Ma Z, Chen J, Li B (2009) Potential impacts of invasive *Spartina alterniflora* on spring bird communities at Chongming Dongtan, a Chinese wetland of international importance [Article]. *Estuarine Coast Shelf Sci* 83(2):211–218. <https://doi.org/10.1016/j.ecss.2009.03.026>
- Gao S, Du Y, Xie W, Gao W, Wang D, Wu X (2014) Environment-ecosystem dynamic processes of *Spartina alterniflora* salt-marshes along the eastern China coastlines [Review]. *Sci China-Earth Sci* 57(11):2567–2586. <https://doi.org/10.1007/s11430-014-4954-9>
- Gao Y, Peng R-H, Ouyang Z-T, Shao C-L, Chen J-Q, Zhang T-T, Guo H-Q, Tang J-W, Zhao F, Zhuang P, Zhao B. (2020) Enhanced Lateral Exchange of Carbon and Nitrogen in a Coastal Wetland With Invasive *Spartina alterniflora* [Article]. *J Geophys Res-Biogeosci* 125(5):e2019JG005459. <https://doi.org/10.1029/2019jg005459>
- Ge Z, Cao H, Zhang L (2013) A process-based grid model for the simulation of range expansion of *Spartina alterniflora* on the coastal saltmarshes in the Yangtze Estuary [Article]. *Ecol Eng* 58:105–112. <https://doi.org/10.1016/j.ecoleng.2013.06.024>
- Gong L, Song Y, Zhao W, Su L, Wu S, Hu Y, Li B, Li X (2025) Evaluation of carbon flux responses to physical and chemical control strategies of *Spartina alterniflora* in the Yangtze River Delta [Article]. *Plant Soil* 514(2):2159–2176. <https://doi.org/10.1007/s11104-025-07510-7>
- Gong Z, Shi L, Jin C, Zhang Q, Zhao K (2021) Long term evolution of tidal flat at the center of Jiangsu Province and the influence of *Spartina alterniflora* growth. *Adv Wat Sci* 32(4):618–626, Article 1001–6791(2021)32:4<618:Jszbct>2.0.Tx;2-a
- Gong L, Hu Y, Li B, Song Y, Li Z, Li T, Li X (2024) Effects of *Spartina alterniflora* eradication project on macrobenthos community structure: A case study of Nanhui tidal flat in Shanghai [Article]. *Chinese J Ecol* 43(9):2892–2900, Article 1000–4890(2024)43:9<2892:Hmcmz>2.0.Tx;2-b.
- Gong L, Hu Y, Song Y, Pan L, Su L, Yu X, Zhao W, Li B, Li X (2026) Evaluating sustainable coastal management strategies following *Spartina alterniflora* eradication: Insights from typhoon-induced hydro-sedimentary responses [Article]. *Ocean Coast Manag* 276:108131. <https://doi.org/10.1016/j.ocecoaman.2026.108131>
- Gu J, Wu J (2023) Blue carbon effects of mangrove restoration in subtropics where *Spartina alterniflora* invaded [Article]. *Ecol Eng* 186:106822. <https://doi.org/10.1016/j.ecoleng.2022.106822>
- Gu W, Li G, Zhao X, Shi B, Zhang W, Ding J (2025) Research on the impact of various vegetation types on wave attenuation along the coast of Zhejiang province:taking *Scirpus mariqueter*, *Spartina alterniflora*, and mangrove ecosystems as examples [Article]. *The Ocean Eng* 43(5):122–131, Article 1005–9865(2025)43:5<122:Zjshab>2.0.Tx;2-3.
- Han Q, Han Q, Wang Y, Liu D (2022) Comparison of macrobenthic communities between the invasive *Spartina alterniflora* and native *Suaeda glauca* habitats in the Yellow River Delta [Article]. *Wetlands Ecol Manag* 30(3):497–511. <https://doi.org/10.1007/s11273-022-09876-2>
- Hou W, Liu H, Xiang Z, Yan H, Li G, Bai J (2026) Divergent soil microbial patterns despite similar TOC levels in *Spartina alterniflora* invasion and early-stage *Scirpus mariqueter* restoration [Article]. *Estuarine Coast Shelf Sci* 335:109857. <https://doi.org/10.1016/j.ecss.2026.109857>
- Hu Y, Zhu W, Ren G, Wu P, Wang J, Zhu W, Xin H, Fu S, Xu H, Ma Y (2026b) Remote Sensing Identification and Mapping of *Spartina Alterniflora* in Chinese Mainland Coastal at 2 m Spatial Resolution [Article]. *Ieee J Select Topics Appl Earth Observ Remote Sens* 19:399–420. <https://doi.org/10.1109/JSTARS.2025.3629118>
- Hu W, Gong C, Wu D, Wang Z, Liu B, Sun C, An S, Lu C (2026) Short-term effects of a tilling-based *Spartina alterniflora* removal project on intertidal macrobenthic communities [Article]. *J Environ Manag* 404:129329. <https://doi.org/10.1016/j.jenvman.2026.129329>
- ISSG. (2020). Species profile: *Spartina alterniflora*. IUCN. <http://www.iucngisd.org/gisd/speciesname/Spartina+alterniflora>
- Jackson M. V, Fuller R. A, Gan X, Li J, Mao D, Melville D. S, Murray N. J, Wang Z, Choi C.-Y (2021) Dual threat of tidal flat loss and invasive *Spartina alterniflora* endanger important shorebird habitat in coastal mainland China [Article]. *J Environ Manag* 278: 111549. <https://doi.org/10.1016/j.jenvman.2020.111549>

- Jiang C, Jia J, Yang Y, Sheng R, Ren L, Ji H, Ye S (2024) Forty years of *Spartina alterniflora* in China: cognitive evolution and governance strategies [Article]. *Acta Ecol Sinica* 44(20):8944–8956, Article 1000–0933(2024)44:20<8944:Hhmczz>2.0.Tx;2–3.
- Jin B, Yan H, Zhang L, Zeng C (2016) Spatial-temporal Variations and their Influence Factors of Soil Organic Carbon under the *Spartina alterniflora* Wetland in China. *Ecol Environ Sci* 25(12):2021–2027, Article 1674–5906(2016)25:12<2021:Zgbhnh>2.0.Tx;2–4
- Jin B, Yan H, Zhang W, Zeng C (2017) Contents and Storages of Various Forms of Nitrogen in Soils of Wetlands in the Min River Estuary under *Spartina alterniflora* Invasion. *Wetland Sci* 15(3):375–384, Article 1672–5948(2017)15:3<375:Hhmcrcq>2.0.Tx;2–5
- Kerr DW, Hogle IB, Ort BS, Thornton WJ (2016) A review of 15 years of *Spartina* management in the San Francisco Estuary [Review]. *Biol Invasions* 18(8):2247–2266. <https://doi.org/10.1007/s10530-016-1178-2>
- Kneib RT, Newell SY, Hermeno ET (1997) Survival, growth and reproduction of the salt-marsh amphipod *Uholorchestia spartinophila* reared on natural diets of senescent and dead *Spartina alterniflora* leaves [Article]. *Mar Biol* 128(3):423–431. <https://doi.org/10.1007/s002270050108>
- Kong W, Lu L, Li X, Li X, Gao X, Wang Y, Xing M, Zhang C, Xu J, Liu H (2025) Effectiveness of herbicides for the control and removal of *Spartina alterniflora* and their molecular response [Article]. *J Appl Oceanograph* 44(4):673–681, Article 2095–4972(2025)44:4<673:Cjgqch>2.0.Tx;2–t.
- Leng Z, Liu J, He C, Wang Z, He S, Du D, Li J (2025) Deposition of sulfur by *Spartina alterniflora* promoted its ecological adaptability in cadmium-polluted coastal wetlands [Article]. *Bioresour Technol* 419:132069. <https://doi.org/10.1016/j.biortech.2025.132069>
- Li J, L, Xu J, Q, Zhang D, F, Et A (2005) Function of *Spartina alterniflora* salt marsh and its eco-economic value in south coast of Hangzhou Bay (in Chinese). *Area Res Dev*(24):58–62.
- Li J, Jin S, Li R, Shao S, Yang W (2024) Characteristics of Changes in Macroben-thos Community Structure after Mowing and Ploughing of *Spartina alterniflora* in LuannanWetland [Article]. *Wetland Sci* 22(3):349–358, Article 1672–5948(2024)22:3<349:Ygfgzl>2.0.Tx;2–6.
- Li Q-W, Gao J-Q, Guo Y, Liang J-F, Yu F-H (2025) Effects of biochar on nitrogen competition between invasive *Spartina alterniflora* and native *Phragmites australis* [Article]. *J Plant Ecol* 18(4):rtaf063. <https://doi.org/10.1093/jpe/rtaf063>
- Li S-H, Ge Z-M, Tan L-S, Xie L-N, Li Y-L (2025) Multiple competitive superiority made a great successful invasion of *Spartina alterniflora* in Eastern China: hints for management [Article]. *J Environ Manag* 389:126287. <https://doi.org/10.1016/j.jenvman.2025.126287>
- Li X, Zhang X, Gao J-Q, Li M-H, Penuelas J, Sardans J, Li Q, Song M, Liang C, Liu H, Zhu J, Yu F-H (2026) *Spartina alterniflora* invasion enhances soil priming effect by altering carbon use efficiency [Article]. *Catena* 267:Article 109972. <https://doi.org/10.1016/j.catena.2026.109972>
- Liu S, Liu W, He C, Hua J, Tao Y, Wang D, Zhao Z, Chen X, Wang F, Liu X (2026) Impact of *Spartina alterniflora* root systems on Fe-bound organic carbon and bacterial communities of soil in Jiuduansha wetland [Article Early Access]. *Plant Soil*. <https://doi.org/10.1007/s11104-026-08394-x>
- Liu J, Su H, Xu J, Huang H, Wang G, Chen J (2017) How does *Spartina alterniflora* Affect the Soil Organic Carbon Pool of Coastal Wetlands in China. *Ecol Environ Sci* 26(6):1085–1092, Article 1674–5906(2017)26:6<1085:Hhmcdz>2.0.Tx;2–y
- Liu N, Yang Y, Tong C, Luo M, Huang J (2023) Soil microbial metabolism limitation and carbon use efficiency in four different wetland habitats of Jiulong River estuary [Article]. *Acta Sci Circumstantiae* 43(6):471–482, Article 0253–2468(2023)43:6<471:Jlhjks>2.0.Tx;2–4.
- Liu W, Tao Y, He P, Liu J, Zhang W (2025) Assessing the impacts of climate change on suitable distribution areas and ecological risks of the invasive grass (*Spartina alterniflora*) in China [Article]. *J Nat Conserv* 87:126985. <https://doi.org/10.1016/j.jnc.2025.126985>
- Long D, Zhao W, Li X, Sun Q, Li J, Lin X (2025) Rhizosphere Effect Enhances Belowground Competition of Coastal Invasive *Spartina alterniflora* With Mangroves [Article]. *Ecol Evol* 15(12):e72565. <https://doi.org/10.1002/ece3.72565>
- Lowe S, Browne M, Boudjelas S, De Poorter M (2000) 100 of the world's worst invasive alien species: A selection from the Global Invasive Species Database. Invasive Species Specialist Group (ISSG), IUCN SSC. Originally published as a lift-out supplement in *Aliens*, 12
- Lu J, Zhang Y (2013) Spatial distribution of an invasive plant *Spartina alterniflora* and its potential as biofuels in China [Article]. *Ecol Eng* 52:175–181. <https://doi.org/10.1016/j.ecoleng.2012.12.107>
- Lyu C, Zhang S, Ren X, Liu M, Leung K-S K, He T, Chen Q, Choi C.-Y. (2023) The effect of *Spartina alterniflora* eradication on waterbirds and benthic organisms [Article]. *Restor Ecol* 31(8). <https://doi.org/10.1111/rec.14023>
- Maricle B R, Lee R W (2002) Aerenchyma development and oxygen transport in the estuarine cordgrasses *Spartina alterniflora* and *S. anglica* [Article]. *Aquat Botan* 74(2):109–120, Article Pii s0304–3770(02)0051–7. [https://doi.org/10.1016/s0304-3770\(02\)0051-7](https://doi.org/10.1016/s0304-3770(02)0051-7)
- Mavrodi O V, Jung C M, Eberly J O, Hendry S V, Namjilsuren S, Biber P D, Indest K J, Mavrodi D V (2018) Rhizosphere Microbial Communities of *Spartina alterniflora* and *Juncus roemerianus* From Restored and Natural Tidal Marshes on Deer Island, Mississippi [Article]. *Front Microbiol* 9: 3049. <https://doi.org/10.3389/fmicb.2018.03049>
- Meng W, Feagin R A, Innocenti R A, Hu B, He M, Li H (2020) Invasion and ecological effects of exotic smooth cordgrass *Spartina alterniflora* in China [Review]. *Ecol Eng* 143: 105670. <https://doi.org/10.1016/j.ecoleng.2019.105670>
- Min Y, Ke Y, Zhuo Z, Qi W, Li J, Li P, Zhao N (2025) Monitoring *Spartina alterniflora* removal dynamics across coastal China using time series Sentinel-1 imagery [Article]. *Remote Sens Environ* 326;114813. <https://doi.org/10.1016/j.rse.2025.114813>
- Mo S, Wu X, Kashif M, Zeng S, Sang Y, Meng C, He S, Jiang C (2025) Effects of *Spartina alterniflora* invasion on carbon fixation and sulfate reduction in a subtropical marine mangrove ecosystem [Article]. *Marine Pollut Bull* 217: 118128. <https://doi.org/10.1016/j.marpolbul.2025.118128>
- Mooney HA, Drake JA (Eds.) (1986) Ecology of biological invasions of North America and Hawaii. *Ecol Stud* 58. Springer. <https://doi.org/10.1007/978-1-4612-4988-7>
- Nellemann C, Corcoran E, Duarte CM, Valdés L, DeYoung, CG, Fonseca LE, Grimsditch GD (2009) Blue carbon: The role of healthy oceans in binding carbon – A rapid response assessment. United Nations Environment Programme, GRID-Arendal. <https://www.grida.no/publications/145>
- Ning ZH, Chen C, Zhang SY, Wang AD, Wang Q, Xie T, Bai JH, Cui BS (2023) Lateral Hydrological Connectivity Driven by Tidal Flooding Regulates Range- Expansion of Invasive *Spartina alterniflora* in Tidal Channel-Salt Marsh Systems [Article]. *J Environ Inf* 41(1):16–26. <https://doi.org/10.3808/jei.202300484>
- Partridge TR (1987) *Spartina* in New Zealand. *NZ J Bot* 25:567–575
- Pastore MA, Megonigal JP, Langley JA (2017) Elevated CO₂ and nitrogen addition accelerate net carbon gain in a brackish marsh [Article]. *Biogeochemistry* 133(1):73–87. <https://doi.org/10.1007/s10533-017-0312-2>
- Qin P, Xie M, Zhong CX (1989) The distribution of 18 metals in the salt marsh of estuary Luoyuan, Fujian (in Chinese). *Mar Sci* 6:23–27
- Qin P, Xie M, Jiang YS, Chung CH (1997) Estimation of the ecological-economic benefits of two *Spartina alterniflora* plantations in North Jiangsu, China [Article]. *Ecol Eng* 8(1):5–17. [https://doi.org/10.1016/s0925-8574\(96\)00227-3](https://doi.org/10.1016/s0925-8574(96)00227-3)
- Rui-Zhao W, Li-Quan Z (2009) Effect of managed waterlogging to control *Spartina alterniflora* on macrobenthic communities. *Acta Ecol Sinica* 29(5):2639–2645, Article 1000–0933(2009)29:5<2639:Swtkcs>2.0.Tx;2–8
- Sengupta S, Mangu V, Sanchez L, Bedre R, Joshi R, Rajasekaran K, Baisakh N (2019) An actin-depolymerizing factor from the halophyte smooth cordgrass, *Spartina alterniflora* (SaADF2), is superior to its rice homolog (OsADF2) in conferring drought and salt tolerance when constitutively overexpressed in rice [Article]. *Plant Biotechnol J* 17(1):188–205. <https://doi.org/10.1111/pbi.12957>
- Song X, Wang F, Hai J, Liu Y, Wang H, Wei S, Chen H (2025) Analyses of decomposition dynamics of organic components in the leaf litter of native plant *Kandelia obovata* and invasive plant *Spartina alterniflora* [Article]. *Xiamen Daxue Xuebao (Ziran Kexue Ban)* 64(3):383–393. <https://doi.org/10.6043/j.issn.0438-0479.202407018>
- Song L, Wang Q, Wang P, Wu J (2023) Benthic bacterial communities and bacteria-environment interactions after *Kandelia obovata* introduction and *Spartina alterniflora* invasion in Yueqing Bay, China [Article]. *Region Stud Marine Sci* 58: 102787. <https://doi.org/10.1016/j.rsma.2022.102787>
- Sui Y, Wei Y, Feng W, Liao D, Yan C, Chen J, Lv L (2025) Impacts of *Spartina alterniflora* Control Measures on Wetland Aquatic Communities:

- eDNA-Based Diversity Responses of Fish and Zooplankton [Article]. *Wetlands* 45(8):123. <https://doi.org/10.1007/s13157-025-02010-x>
- Sun P, Wu Y, Zhu P, Wang J, Yu X, Guo W (2025) *Spartina alterniflora* invasion significantly alters the assembly and structure of soil bacterial communities in the Yellow River Delta [Article]. *Front Microbiol* 16:1525632. <https://doi.org/10.3389/fmicb.2025.1525632>
- Tang C, Luo L, Shi J, Liu X, Xian W, Wang J (2026) A potentially overlooked benefit of *Spartina alterniflora* invasion: tidal transport of root-exuded active sugars enhancing soil carbon accumulation via microbial-mediated processes in adjacent mudflats [Article]. *J Soils Sediments* 26(3):72. <https://doi.org/10.1007/s11368-026-04292-y>
- Tyler AC, Grosholz ED, Civile JC, Lambrinos JG (2003) Changes in carbon and nitrogen cycling following *Spartina alterniflora* invasion of Pacific estuaries [Meeting]. *Ecological Society of America Annual Meeting Abstracts* 88:339–339
- Wan S, Qin P, Liu J, Zhou H (2009) The positive and negative effects of exotic *Spartina alterniflora* in China [Article]. *Ecol Eng* 35(4):444–452. <https://doi.org/10.1016/j.ecoleng.2008.05.020>
- Wang Q, An S-Q, Ma Z-J, Zhao B, Chen J-K, Li B (2006a) Invasive *Spartina alterniflora*: biology, ecology and management [Review]. *Acta Phytotaxonomica Sinica* 44(5):559–588. <https://doi.org/10.1360/aps06044>
- Wang Q, Wang CH, Zhao B, Ma ZJ, Luo YQ, Chen JK, Li B (2006b) Effects of growing conditions on the growth of and interactions between salt marsh plants: Implications for invasibility of habitats [Article]. *Biol Invasions* 8(7):1547–1560. <https://doi.org/10.1007/s10530-005-5846-x>
- Wang M, Gao X, Wang W (2014) Differences in burrow morphology of crabs between *Spartina alterniflora* marsh and mangrove habitats [Article]. *Ecol Eng* 69:213–219. <https://doi.org/10.1016/j.ecoleng.2014.03.096>
- Wang L, Yuan J, Wang Y, Butterly CR, Tong D, Zhou B, Li X, Zhang H (2021) Effects of Exotic *Spartina alterniflora* Invasion on Soil Phosphorus and Carbon Pools and Associated Soil Microbial Community Composition in Coastal Wetlands [Article]. *ACS Omega* 6(8):5730–5738. <https://doi.org/10.1021/acsomega.0c06161>
- Wang Y, Kong J, Gu S, Huang B, Sun P (2024e) Dynamics of benthic micro-eukaryotic communities in a mangrove wetland invaded by *Spartina alterniflora*: Effects of vegetation, seasonality, and sediment depth [Journal Article]. *The Science of the Total Environment* 916:170231–170231. <https://doi.org/10.1016/j.scitotenv.2024.170231>
- Wang D, Labra FA, Yang H, Hu Y, Zhao Z, Zhou W, Yuan L (2025a) Restoration of native saltmarshes enhances carbon sequestration and mitigates warming effects following *Spartina alterniflora* removal [Article]. *J Appl Ecol* 62(8):2005–2017. <https://doi.org/10.1111/1365-2664.70096>
- Wang F, Zhang N, Yang S, Li Y, Yang L, Cao W (2024) Source and stability of soil organic carbon jointly regulate soil carbon pool, but source alteration is more effective in mangrove ecosystem following *Spartina alterniflora* invasion. *Catena Article* 107681. <https://doi.org/10.1016/j.catena.2023.107681>
- Wang J, Lin X, An X, Liu S, Wei X, Zhou T, Li Q, Chen Q, Liu X (2024) Mangrove afforestation as an ecological control of invasive *Spartina alterniflora* fl ora affects rhizosphere soil physicochemical properties and bacterial community in a subtropical tidal estuarine wetland [Article]. *PEERJ* 12:E18291. <https://doi.org/10.7717/peerj.18291>
- Wang K, Wang S, Zhang X, Wang W, Wang X, Kong F, Xi M (2024) The amelioration and improvement effects of modified biochar derived from *Spartina alterniflora* on coastal wetland soil and Suaeda salsa growth [Article]. *Environ Res* 240:117426. <https://doi.org/10.1016/j.envres.2023.117426>
- Wang X-Y, Sun Z-G, Chen B-B, Wu H-H, Zhang D-Y (2024) Ex situ decomposition and phosphorus release characteristics of *Spartina alterniflora* litter in Minjiang estuary [Article]. *Chinese J Plant Ecol* 48(7):844–857. <https://doi.org/10.17521/cjpe.2022.0489>
- Wang J, Chen X, Wang Y, Chen Y, Wu H, Wittingham S S, Kirwan M L, Zhang Y, Liu W, Zhang Y (2025) Life-history plasticity of intertidal salt marsh in response to sea level rise: Salinity and inundation modulate size-dependent flowering of *Spartina alterniflora* [Article]. *Ecol Indic* 177:113790. <https://doi.org/10.1016/j.ecolind.2025.113790>
- Wang L, Li Y, Hei J, Wang W, Sardans J, Zhang Z, Zeng F, Ge M, Liao Y, Fang Y, Vancov T, Gan J, Song Z, Zhang W, Penuelas J. (2025) Impacts of *Spartina alterniflora* invasion on soil carbon components of particulate and mineral-associated organic matter and soil organic matter mineralization in estuarine wetlands [Article]. *Appl Soil Ecol* 206:105857. <https://doi.org/10.1016/j.apsoil.2024.105857>
- Wang R, Su Y, Sun X, Wang M, Feng M (2025) Rapid and automated mapping method of *Spartina alterniflora* combines tidal imagery and phenological characteristics [Article]. *Environ Monitor Assess* 197(10):1136. <https://doi.org/10.1007/s10661-025-14580-8>
- Wang Y, Li N, Zhang Y, Guo C, Xie L, Li Y, Guo H (2025) Nitrogen Addition Reduces Negative Plant-Soil Feedback in Invasive *Spartina alterniflora*: Preliminary Findings from a Mesocosm Experiment [Article]. *Agronomy-Basel* 16(1):86. <https://doi.org/10.3390/agronomy16010086>
- Wei S, Zhu Z, Wang S (2024) Spatio-temporal dynamics of net primary productivity and the economic value of *Spartina alterniflora* in the coastal regions of China [Article]. *Sci Total Environ* 953:176099. <https://doi.org/10.1016/j.scitotenv.2024.176099>
- Weston NB, Neubauer SC, Velinsky DJ, Vile MA (2014) Net ecosystem carbon exchange and the greenhouse gas balance of tidal marshes along an estuarine salinity gradient [Article]. *Biogeochemistry* 120(1–3):163–189. <https://doi.org/10.1007/s10533-014-9989-7>
- Wilburn B P, Raper K, Raposa K B, Gray A B, Mozdzer T J, Watson E. B (2024) Promoting success in thin layer sediment placement: effects of sediment grain size and amendments on salt marsh plant growth and greenhouse gas exchange [Article]. *Restor Ecol* 32(5). <https://doi.org/10.1111/rec.14141>
- Williams BL, Johnson DS (2021) Role of ecological interactions in saltmarsh geomorphic processes [Article]. *Mar Ecol Prog Ser* 658:149–161. <https://doi.org/10.3354/meps13554>
- Wu Y, Xu B, Sun D, Jiang S, Li J, Wang Y, Wu Y (2025) Higher dynamic and turnover yet enhanced organic carbon burial in *Spartina alterniflora* sediments: A case study in a temperate coastal wetland [Article]. *Catena* 249:108701. <https://doi.org/10.1016/j.catena.2025.108701>
- Wu Y, Sun D, Jiang S, Xu B, Liu J, Li J, Wang Y, Wu Y (2026) Distinct mowing effects on organic carbon storage and dynamics in *Spartina alterniflora* and *Phragmites australis* sediments [Article]. *Ocean Coast Manag* 274:108078. <https://doi.org/10.1016/j.ocecoaman.2025.108078>
- Wu Z, Li Y, Wang W, Liu X, Hou N, Hei J, Gan J, Song Z, Fang Y, Sardans J, Penuelas J (2026) Impacts of *Spartina alterniflora* invasion on fractions and fungal communities of mineral-associated organic carbon in subtropical coastal wetlands of China [Article]. *Catena* 266:109956. <https://doi.org/10.1016/j.catena.2026.109956>
- Xie S, Li X (2024) Effects of *Spartina alterniflora* Invasion on Soil Carbon Budget in Coastal Wetlands of China [Article]. *Ecol Environ Sci* 33(10):1516–1524, Article 1674–5906(2024)33:10<1516:Hhmcrcq>2.0.Tx;2-t.
- Xu X, Fu G, Zou X, Ge C, Zhao Y (2017) Diurnal variations of carbon dioxide, methane, and nitrous oxide fluxes from invasive *Spartina alterniflora* dominated coastal wetland in northern Jiangsu Province [Article]. *Acta Oceanol Sin* 36(4):105–113. <https://doi.org/10.1007/s13131-017-1015-1>
- Yan D, Wang S, Song P, Zhang Y, Huang Y, Wang Y, Zhu W, Wu L (2025) Research hotspots and trends in *Spartina alterniflora* studies (1970–2023): A bibliometric analysis [Review]. *Marine Pollut Bull* 221:118483. <https://doi.org/10.1016/j.marpolbul.2025.118483>
- Yang R-M, Yang F (2020) Impacts of *Spartina alterniflora* invasion on soil inorganic carbon in coastal wetlands in China [Article]. *Soil Sci Soc Am J* 84(3):844–855. <https://doi.org/10.1002/saj2.20073>
- Yang W, Zhao H, Leng X, Cheng X, An S (2017) Soil organic carbon and nitrogen dynamics following *Spartina alterniflora* invasion in a coastal wetland of eastern China [Article]. *CATENA* 156:281–289. <https://doi.org/10.1016/j.catena.2017.03.021>
- Yang J, Cui L, Zhuo Z, Tian J, Li P, Min Y, Ke Y (2025) Rapid change in structural stability of tidal flats in response to large-scale eradication of *Spartina alterniflora*: evidence from dense time-series PlanetScope satellite imagery [Article]. *J Environ Manag* 395:127652. <https://doi.org/10.1016/j.jenvman.2025.127652>
- Yang L, Pan R, Wang S, Zhu Z, Li H, Yang R, Sun X, Ge B (2025) Macrofaunal biodiversity and trophic structure varied in response to changing environmental properties along the *Spartina alterniflora* invasion stages [Article]. *Marine Pollut Bull* 214:117756. <https://doi.org/10.1016/j.marpolbul.2025.117756>
- Yao Y, Song Y, Su P, Wang J, Miao C, Luo Y, Sun Q, Wang J, Zhang G, Bu N, Li Z (2023) Asymmetric responses of functional microbes in methane and nitrous oxide emissions to plant invasion: A meta-analysis [Article]. *Soil Biol Biochem* 178:108931. <https://doi.org/10.1016/j.soilbio.2022.108931>

- Zhang D, Hu Y, Liu M, Chang Y, Yan X, Bu R, Zhao D, Li Z (2017) Introduction and Spread of an Exotic Plant, *Spartina alterniflora*, Along Coastal Marshes of China [Article]. *Wetlands* 37(6):1181–1193. <https://doi.org/10.1007/s13157-017-0950-0>
- Zhang Y, Meng H, Wang Y, He Q (2018) Herbivory enhances the resistance of mangrove forest to cordgrass invasion [Article]. *Ecology* 99(6):1382–1390. <https://doi.org/10.1002/ecy.2233>
- Zhang J, Mao D, Liu J, Chen Y, Kirwan M, Sanders C, Zhou J, Lu Z, Qin G, Huang X, Li H, Yan H, Jiao N, Su J, Wang F (2024b) *Spartina alterniflora* invasion benefits blue carbon sequestration in China [Article]. *SCIENCE BULLETIN* 69(12):1991–2000. <https://doi.org/10.1016/j.scib.2024.04.049>
- Zhang H, Zhen Y, Wu F, Li Y, Zhang Y (2020) Relationship between habitat quality change and the expansion of *Spartina alterniflora* in the coastal area: Taking Yancheng National Nature Reserve in Jiangsu Province as an example. *Resour Sci* 42(5):1004–1014, Article 1007–7588(2020)42:5<1004:Bhdsj>2.0.Tx;2-f
- Zhang G, Bai J, Zhao Q, Jia J, Wang X, Wang W, Wang X (2021) Soil carbon storage and carbon sources under different *Spartina alterniflora* invasion periods in a salt marsh ecosystem [Article]. *Catena* 196:104831. <https://doi.org/10.1016/j.catena.2020.104831>
- Zhang M, Schwarz C, Lin W, Naing H, Cai H, Zhu Z (2023) A new perspective on the impacts of *Spartina alterniflora* invasion on Chinese wetlands in the context of climate change: A case study of the Jiuduansha Shoals, Yangtze Estuary [Article]. *Sci Total Environ* 868:161477. <https://doi.org/10.1016/j.scitotenv.2023.161477>
- Zhang X, Xiao X, Wang X, Xu X, Qiu S, Pan L, Ma J, Ju R, Wu J, Li B (2023) Continual expansion of *Spartina alterniflora* in the temperate and subtropical coastal zones of China during 1985–2020 [Article]. *Int J Appl Earth Observ Geoinform* 117:103192. <https://doi.org/10.1016/j.jag.2023.103192>
- Zhang G, Bai J, Tebbe C C, Huang L, Jia J, Wang W, Wang X, Zhao Q, Wen L, Kong F, Xi M, He Q (2024) Habitat-specific responses of soil organic matter decomposition to *Spartina alterniflora* invasion along China's coast [Article]. *Ecol Appl* 34(1):e2741. <https://doi.org/10.1002/eap.2741>
- Zhang N, Luo L, Xiang H, Zhen J, Li A, Wang Z, Mao D (2025) A Review of Remote Sensing on *Spartina alterniflora*: Status, Challenge, and Direction [Review]. *Remote Sens* 17(24):3951. <https://doi.org/10.3390/rs17243951>
- Zhang X, Ma B, Zhang Q, Wang Q, Yang X, Fei J, Zhang S, Zhang F, Zhang Y, Xu S (2025) Impacts of *Spartina alterniflora* invasion on C-cycling functional genes and microbial communities in salt marsh wetlands during cold season [Article]. *Ecol Process* 14(1):53. <https://doi.org/10.1186/s13717-025-00592-2>
- Zhang Q, Mao D, Man W, Li F, Zhang Y, Wu F, Kou C, Yang R, He J, Yin X, Liu M (2026) *Spartina alterniflora* invasion-induced soil organic carbon content changes: An assessment by time-series remote sensing and machine learning [Article]. *Geoderma* 465:117650. <https://doi.org/10.1016/j.geoderma.2025.117650>
- Zhao W, Li X, Xue L, Lin S, Ma Y, Su L, Li Z, Gong L, Yan Z, Macreadie PI (2023) Mapping trade-offs among key ecosystem functions in tidal marsh to inform spatial management policy for exotic *Spartina alterniflora* [Journal Article]. *J Environ Manage* 348:119216–119216. <https://doi.org/10.1016/j.jenvman.2023.119216>
- Zhao Z, Yuan L, Li W, Zhu X, Pan J, Chen Y, Zhang L (2018) Effects of habitat heterogeneity and ortet density on the invasiveness of *Spartina alterniflora*. *Acta Ecol Sinica* 38(18):6632–6641, Article 1000–0933(2018)38:18<6632:Sjyzxj>2.0.Tx;2–8
- Zhao Q, Xie B, Bi X, Cao Q, Han G, Song W, Wang X (2025) *Spartina alterniflora* invasion impacts soil organic carbon sequestration and stability in salt marsh wetlands [Article]. *Environ Technol Inn* 39:104320. <https://doi.org/10.1016/j.eti.2025.104320>
- Zhou H-X, Liu J-E, Qin P (2009) Impacts of an alien species (*Spartina alterniflora*) on the macrobenthos community of Jiangsu coastal inter-tidal ecosystem [Article]. *Ecol Eng* 35(4):521–528. <https://doi.org/10.1016/j.ecoleng.2008.06.007>
- Zhou B, Xu M, Tian J, Huang Y, Song J, Zhu L, Zhu X, Qu X, Zhang L, Li X, Gong H (2024) Mapping the invasive *Spartina alterniflora* in sub-meter level with improved phenological spectral features and deep learning method [Review]. *Int J Digit Earth* 17(1):2434634. <https://doi.org/10.1080/17538947.2024.2434634>
- Zhou B, Xu M, Tian J, Jia M, Mao D, Cheng K, Zhu X, Jiang H, Song J, Ke Y, Zhang Z, Huang Y, Wei M, Zhu L, Li X, Gong H (2025) National-scale sub-meter mapping of *Spartina alterniflora* in mainland China 2020 [Article Data Paper]. *Earth Syst Sci Data* 17(11):6601–6620. <https://doi.org/10.5194/essd-17-6601-2025>
- Zhu X, Meng L, Zhang Y, Weng Q, Morris J (2019) Tidal and Meteorological Influences on the Growth of Invasive *Spartina alterniflora*: Evidence from UAV Remote Sensing [Article]. *Remote Sens* 11(10):1208. <https://doi.org/10.3390/rs11101208>
- Zhu K-H, Ge Z-M, Huang Y, Zhao L-H, Li Z-F, Zhao W, Chen H-Y, Zhang D, Cheng H-F, Zhang W, Xin P (2025) Biogeomorphology and carbon sequestration in a coastal shoal invaded by *Spartina alterniflora* in the Yangtze Estuary: 22-year simulation for management implication [Article]. *Ecol Eng* 220:107756. <https://doi.org/10.1016/j.ecoleng.2025.107756>
- Zhu X, Qin Z, Liu W, Kirwan M L, Lu H, Lee S Y, Dai M (2025) Coastal Restoration May Not Necessarily Enhance Blue Carbon Sink [Article]. *Geophys Res Lett* 52(11):e2025GL114614. <https://doi.org/10.1029/2025GL114614>
- Zhu Y, Wu Y, Xue J (2025) A review of the impacts of *Spartina alterniflora* invasion and management on bird habitats in coastal salt marsh wetlands [Review]. *J Nanjing Forestry University. Natural Sciences Edition* 49(6), 281–290, Article 1000–2006(2025)49:6<281:Hhmcrcq>2.0.Tx;2-m.
- Zhu H L, Gan K Y, Tam N F Y, Chen Z T, Li F L, Xu S J L, Peng D, Zhou H C, Lee, F W F (2026) Comparison of meiofauna, particularly free-living marine nematode community in interspersed habitats of mangroves and *Spartina alterniflora* in Zhanjiang, China [Article]. *Mar Pollut Bull* 223:118943. <https://doi.org/10.1016/j.marpolbul.2025.118943>